

School of Computing

MSc Project Report

International Postgraduate Student Enrolment in Scottish Universities: A Data-Driven Analysis with a Focus on Currency Fluctuations

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I confirm that the work contained in this Honours project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

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Abstract

This study examines the relationship between currency fluctuations and international postgraduate student enrolment at Scottish universities from 2014 to 2022. Using a data-driven approach, incorporating statistical analysis, machine learning, and time-series forecasting, the research demonstrates that significant currency depreciations, such as those observed in Nigeria (-0.58 correlation) and Ghana (-0.23), correlate with declining student numbers. Conversely, economic stability in countries like China (0.05 correlation) and India (0.09) supports consistent enrolment, indicating a weaker influence of currency fluctuations.

Additionally, visa policy changes, including the introduction of the Graduate Route in 2021 (linked to a 15% enrolment increase) and the 2024 restrictions on dependents (predicted to decrease enrolment by 10%), significantly impact student decisions. The study quantifies the influence of financial variables using techniques like ARIMA modelling (0.93 R-squared) and Principal Component Analysis (explaining 55.87% of variance).

Results highlight that while tuition and visa fees are important, currency fluctuations have a more substantial impact, particularly for students from economically volatile regions, with a 10% depreciation in the Nigerian Naira, for example, associated with a 5% enrolment drop. This research provides actionable insights for universities to adapt recruitment strategies and financial planning. The study concludes with recommendations for policy adjustments and future research, emphasising a nuanced understanding of how macroeconomic factors shape international student mobility within Scotland's higher education sector.

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Chapter 1: Introduction

1.1 Background

Scottish higher education institutions have recently observed a concerning trend: a significant decline in international postgraduate student enrolment. This decline necessitates a critical re-evaluation and adaptation of recruitment and support strategies to attract and retain this vital demographic. International students are integral to the higher education sector (20% of the student population) (Pinna et al., 2018; Mewett & Sawyer, 2016), contributing not only through tuition fees but also by enriching the academic and cultural diversity of universities.

However, the global landscape, shaped by factors like Brexit, shifting visa policies, and economic instability, has introduced volatility in international student numbers across the UK. For instance, while the 2021/2022 academic year witnessed a 12.3% surge in international enrolments (Study in the UK, 2023), primarily driven by students from China, India, and Nigeria, subsequent years have experienced a downturn. This decline is further corroborated by a 5% decrease in student visas granted in 2023 compared to the record high of 484,000 in 2022, with postgraduate entrants particularly impacted in early 2024 (ICEF, 2024).

Currency instability has emerged as a critical factor influencing these trends, significantly impacting the financial feasibility of studying abroad. To illustrate, consider the experience of a Nigerian student embarking on postgraduate studies in Scotland during the 2023/2024 academic year. When initially planning their education, the exchange rate was ₦600/£1. However, within 6-8 months, during the visa application process, the rate surged to ₦1,100/£1, eventually escalating to ₦2,000/£1. This drastic shift forced the student to expend significantly more than initially budgeted, highlighting how such volatility can deter prospective students from pursuing education in the UK.

These challenges emphasise the need for a data-driven approach to understand the complex interplay between currency fluctuations and international postgraduate student

enrolment in Scottish universities. Sir Arthur Conan Doyle states in his novel , Sherlock Holmes “aptly stated, ***"It is a capital mistake to theorise before one has data."*** By analysing historical trends and identifying correlations, universities can develop more accurate forecasting models to inform strategic planning, budgeting, and targeted recruitment.

1.1.1 International Student Mobility Trends: Highlighting the Importance of Postgraduates in Scotland

The UK has seen a significant increase in international student numbers from 450,835 in 2016-2017 to 679,970 in the academic year 2021/2022 (HESA, 2023). However, despite this growth, the UK's global market share of international students has declined from 11% to 8%, losing ground to competitors like Australia, Canada, and China (British Council, 2023). This decline is largely attributed to political changes since 2013, including Brexit, stricter immigration legislation, and visa policies, as well as the perception that the UK has become less welcoming to international students (Aisling, 2023). The COVID-19 pandemic further exacerbated this trend, leading to increased departures among international students who arrived during the pandemic, highlighting the fragility of international student mobility in the face of global disruptions.

As for Scotland, international students play a critical role in the High Education sector, being close to a quarter of the total student population by sector for higher education institutions, which is significantly higher, compared to 21 % across the UK. The concentration of international students in Scotland shows that the country has managed to attract talented and ambitious individuals, which provides the nation with significant economic and cultural benefits (Scottish Government, 2021). The growth in international student numbers in Scottish HEIs has been particularly strong among non-EU students, with notable increases from countries such as China, Nigeria, India, and Pakistan. This trend can be attributed to targeted enrolment strategies, enhanced scholarship opportunities, discounts, and Scotland's reputation as a welcoming and high-quality education destination. Moreover, the emphasis on postgraduate students has been

especially important, as they tend to contribute more significantly to research output, innovation, and the overall academic environment, further solidifying Scotland's position in the competitive global education market (Scottish Government, 2021).

1.1.2 Economic Contribution of International Students to the Scottish Government

The Scottish economy highly benefits from international students. These students provide the economy with a considerable sum through expenditure on tuition rates, accommodations, travel, and daily expenses, which in turn, contributes to boosting the local economy and sustaining industries facilitated by living, like housing, hospitality, and retail (Ghann & Abraham, 2020). Furthermore, the cultural diversity introduced by their presence also impacts the Scottish economy by enriching the learning environment of both local and international students, as the perception of this being an international learning arena entices domestic students to apply, even in areas that do not have a markedly high number of domestic applicants (Meharg & Craighill, 2014).

In 2019//2020 academic year, international graduate students alone contributed £4.4 billion to the Scottish economy, which entails the tuition rates, the cost of living, and the creation of jobs as well (Scottish Government, 2021). Moreover, 42% of all these postgraduates continued to remain in Scotland, further boosting the long-term economy and growth of the nation. As such, with international education now being regarded as an export sector, it is a social necessity for Scotland, as it prefers to move towards economic resilience and a global position. (Davey, 2023; Shkoler & Rabenu, 2022).

1.1.3 Economic Volatility in Emerging Markets and Its Impact on International Students Decision Making.

Economic instability in emerging markets, reflected in varying GDP growth and inflation, heavily impacts international students' choices, particularly when it comes to the cost of studying abroad. In West Africa, countries like Nigeria and Ghana face the dual challenge of GDP growth paired with high inflation, which erodes purchasing power and makes education in the UK increasingly unattainable for many families (Edafe, 2024; Iwedi, 2017; Geiger et al., 2019). Similarly, in Asia, strong economies like China and India have seen a steady flow of students to the UK, while nations with weaker economic performance, such as Pakistan and Bangladesh, struggle with high inflation and slower GDP growth, limiting their capacity to send students abroad (Qi et al., 2022; World Bank, 2023). Currency volatility compounds these challenges, as exchange rate fluctuations further inflate the costs of tuition and living expenses, creating a divide where students from more stable economies can afford international education, while those from turbulent regions face significant financial barriers (Kirloskar & Inamdar, 2021).

This economic instability has been shown to directly influence international student enrolment and retention. For instance, Breem et al. (2019) found that a 10% depreciation in the Indian rupee correlates with a 4.5% decrease in the number of Indian students studying in the UK. Similarly, the ongoing devaluation of West African currencies, including the Nigerian naira, has led to a notable decline in student enrolment in higher education institutions (HEIs) abroad (ICEF Monitor, 2024). This not only reduces new enrolments but also impacts current postgraduate students, many of whom struggle to afford the remainder of their tuition fees, leading to disruptions in their studies (Ekugo, 2024).

Furthermore, economic instability also affects tuition fee sensitivity, visa policies, and study destination choices. For example, the removal of post-study work visas in the UK resulted in a significant drop in international student numbers from India, with a 44% decrease in sponsored study visas in the first quarter of 2024 compared to the previous year (Aisling, 2023; Ekugo, 2024). This trend emphasises how policy changes, perceived

economic opportunities post-graduation, and financial realities in students' home countries converge to shape their decisions. Research by the Migration Advisory Committee (MAC) (2018) supports this, indicating that students from economically volatile regions are more likely to choose destinations offering favourable post-study work options or scholarships. These findings highlight the need for higher education institutions to conduct data-driven analyses to understand the correlation between economic conditions and enrolment patterns, enabling them to develop targeted strategies that mitigate the impact of economic volatility on prospective students and attract international students in an increasingly unstable global economic environment.

1.2 Research Aims and Objectives

This research investigates how fluctuations in emerging market currencies impact international postgraduate student enrolment in Scotland. The study analyses the interaction of student demographics (country of origin, mode of study) and currency fluctuations on enrolment rates between 2014-2022. Time permitting, a forecasting model will be developed to predict future enrolment based on currency changes, aiding universities in strategic planning. The objectives of the project are: -

1. Determine the strength and direction of the correlation between currency fluctuations and enrolment of international students in Scottish Universities.
2. Identify specific emerging market currencies that have the most significant impact on enrolment decisions.
3. Provide actionable insights for universities to optimise the recruitment and financial methods associated with currency volatility for their international student populations, to increase the potential of enrolment.
4. Investigate if certain student groups are more vulnerable to financial pressures caused by currency fluctuations when deciding to enrol.
5. Identify demographic factors that could exacerbate or buffer the impact of currency fluctuations on the enrolment of international students.

6. Develop a more nuanced understanding of the enrolment patterns by considering individual student characteristics and financial backgrounds about currency volatility.

1.3 Research Questions

The following questions aim to achieve the goals and objective of this project-

1. What is the strength and direction of the correlation between currency fluctuations and enrolment rates in international students?
2. Which emerging market currencies have the most significant impact on international student enrolment rates?
3. How do the fluctuations in significant currencies impact postgraduate international student enrolment numbers?
4. Can a predictive forecast model accurately predict international postgraduate enrolment numbers based on currency volatility and other factors?
5. Are students from specific countries more likely to be affected by currency fluctuations when deciding to study in Scotland?

1.4 Significance of the Research

This research has important implications for Scottish universities, international students, policy makers, and researchers. Scottish universities will be able to more accurately determine the impact of currency exchange rate on enrolment while enhancing their budgeting strategies and helping international students with better financial support. While relying on the results of this study, international students are educated to make more rational financial decisions when they consider the exchange rate in relation to home country and its implications. Further, policy makers can rely on the current research to develop and implement effective policies in the areas of

international student recruitment, financial aid, and tuition fees to boost the position of Scotland on a global scale. The scientific contribution of this study is manifested in the use of data mining and machine learning techniques to find the hidden patterns and relationships. Thus, further researchers in this field will be able to conduct new studies with the help of the methodology developed in the current research.

1.5 Research Structure

This section demonstrates an overview of the main chapters of this dissertation.

Chapter 2: Provides a comprehensive review of economic theories, it explores existing empirical studies, and identifies gaps in the literature. Data science methods used for enrolment forecasting are also discussed, highlighting the strengths and weaknesses of different approaches. The chapter then summarises key findings, outlining the previous contributions, and identifying specific gaps this research aims to address.

Chapter 3: Details the methodological framework and use of secondary data from sources (HESA, IMF, Oanda etc) to examine how currency fluctuations impact international postgraduate student enrolment in Scotland. Statistical and machine learning techniques analyse the relationship between student demographics, currency volatility, and enrolment, with careful attention paid to ethical considerations surrounding data privacy, potential biases, and responsible use of predictive models.

Chapter 4: Presents research findings, starting with a descriptive analysis of enrolment trends, exchange rates, and tuition fees. The development and evaluation of a predictive model are detailed, including its performance and the factors influencing enrolment. Forecasting scenarios assess the impact of exchange rate and policy changes. The chapter concludes by interpreting the results, discussing their implications for Scottish universities, and acknowledging the study's strengths and limitations.

Chapter 5: Summarises the research outcomes and suggests practical recommendations for universities which include targeted recruitment, flexible tuition

pricing, and proactive financial planning. This chapter also discusses policy recommendations highlighting the need for supportive measures to attract and retain international students. The chapter concludes by acknowledging limitations and suggesting future research avenues.

Chapter 2: Literature Review

2.1 International Student Mobility and Enrolment

2.1.1 Theories and Models of Student Enrolment

The international student decision-making process encompasses an array of theories and models that offer different contradictory and complimentary views on what drives the patterns of student mobility. For students, it's either an option to study closer to home, where the culture and language are alike or to seek change in a foreign academic environment and society. In other words, the foreign students' enrolment decision depends on a mixture of personal preferences, circumstances, and opportunities within the origin and destination countries.

Theory of Planned Behaviour (TPB) by Ajzen is one of the first models in forecasting a student's intention to attend university. This theory models behavioural intentions using attitudes, individual norms, and perceived behavioural control (Ajzen, 1991). This model has been expanded to incorporate characteristics like academic self-concept, which improves its predictive capacity in understanding students' enrolling intentions (Cooper, 2016). In contrast, the **Spady-Tinto** model focuses on individual traits and institutional support, highlighting academic integration and social systems within educational institutions to explain student persistence and dropout rates (Hadjar A. et al. 2022). While both models acknowledge the relevance of personal and societal variables, their emphasis differs. TPB focuses on pre-enrolment goals, while Spady-Tinto discusses post-enrolment persistence.

Tiernan Aisling's research on foreign students who require visas introduces critical viewpoints on the institutional and systemic barriers met by students, exploring concepts like 'stranger encounters' and 'cruel optimism' within the UK's neo-liberalised higher education system (Aisling, 2023). This work highlights the instability and discrimination experienced by international students, an area not focused in the TPB model. Similarly,

Scotland's International Education Strategy (SIES) aligns with TPB by emphasising perceived control and positive attitudes towards study abroad, while addressing broader socio-economic factors, including the impacts of Brexit and changing immigration policies (Scottish Government, 2024). These broader considerations align with reviews of TPB model for its limited attention to societal influences.

Observed data on postgraduate education in the UK draws attention to the fluctuations in international student numbers due to economic crises, policy changes, and visa regulations (House G, 2020). This supports the argument for dynamic models such as Spady-Tinto Model that integrate both individual intentions and external factors.

The increasing international student migration at 186% between 1998 and 2018, can be attributed to the internationalisation of higher education, rise of the middle class in non-Western countries, and the perception of international education as a strategy for social distinction (Weber & Mol, 2023). The "**Student Migration Transition (SMT)**" model suggests that international student migration is closely linked to a country's economic development, suggesting that as economies develop and the middle class expands, the demand for international education rises (Weber & Mol, 2023).

The "**Integration**" model examines the experiences of international students in host countries, highlighting the obscured lines between students as transitional visitors and possible immigrants, as student visas often go before longer-term statuses (Yang, P., 2017). Furthermore, broader theories like **neoclassical economics**, **the new economics of migration**, and **world systems theory** offer insights into the macro-level factors and micro-level decision-making processes influencing international student mobility (Massey, S. D. et al., 1993).

International student mobility has also been explored through models like **migration transition theory**, **network analysis**, and **human capital theory**. Migration transition theory suggests that as countries advance, they initially send more students abroad but eventually become net recipients of international students (De Haas et al., 2020). Network analysis involves the role of social and institutional ties in facilitating student mobility, by highlighting the significance of existing academic networks and regional hubs (Barnett et al., 2016). Human capital theory views international student mobility as an

investment in personal development, driven by prospects of higher earnings and better career opportunities (Beine et al., 2014).

As much as these theories complement each other they are not without complications. While the migration transition theory suggests a shift from emigration to immigration with development, network analysis argues that established academic centres continue to lead regardless of source countries' development levels. Additionally, human capital theory's focus on economic returns may overlook the social and cultural factors emphasised by network analysis.

2.1.2 Push-Pull Factors in International Student Mobility

The push-pull framework is generally applied to students' motivations to study abroad. Push factors which work in the home country and pull factors which work in the host country are thought to have an impact on the students' migration decisions (Cheng et al., 2020).

While the push factors in the student's home country may include such aspects as political instability, economic stagnation, or decline lack of opportunities, and other similar circumstances, the pull factors include the reputation of the host countries educational institutions, scholarships and other financial prompts, value of qualifications, and intercultural opportunities (Aisling, 2023; Herzog, 2005; Maringe & Carter, 2007).

However, studying abroad also presents challenges such as adapting to new environments and the high cost of living, which can hinder students' desire for cultural immersion (Aisling, 2023). Stringent visa regulations and the risk of denial also act as deterrents.

Understanding the interplay of push and pull factors is crucial for policymakers and educational institutions to attract and support international students. This can be achieved through enhanced support services, strategic marketing, and inclusive

immigration policies (Mazzarol & Soutar, 2002). Addressing push factors requires international collaboration to improve educational standards globally.

2.2 The Impact of Economic Factors on International Student Mobility

2.2.1 Currency Fluctuations and Student Decision-Making

The globalisation of higher education has significantly increased the movement of international students (Pawar et al., 2020). However, the COVID-19 pandemic has imposed unprecedented challenges, such as travel restrictions and border closures, leading many students to alter or cancel their plans for overseas education. This decline in international student numbers has severely impacted the financial stability of higher education institutions, especially in Anglophone countries that heavily depend on tuition fees from international students (Mok et al, 2022).

One major factor affecting international student mobility is the economy, particularly currency fluctuations in emerging markets. The pandemic has exacerbated global economic instability, making currency volatility a pressing issue for prospective international students who must consider tuition costs, living expenses, and the value of their home currency (Chevalier, 2022). For instance, the depreciation of the Nigerian Naira, which fell from ₦1/\$1 in 2014 to ₦1/\$0.60 in August 2024, has made studying abroad prohibitively expensive for many Nigerian students. This currency depreciation correlates with a significant drop in international student numbers from Nigeria to the UK by over 20% in the first quarter of 2024 compared to the same period in 2019. Similarly, Sri Lanka's economic crisis has caused a severe foreign currency shortage, making it nearly impossible for Sri Lankan students to afford overseas education (Mok et al., 2022). The Ghanaian Cedi's depreciation has likewise challenged Ghanaian students' ability to study abroad (Hyams–Ssekasi et al., 2014).

Conversely, a strengthening local currency can make studying abroad more feasible by reducing the relative cost of tuition and living expenses. Some emerging markets have responded by introducing scholarship programs and financial aid to mitigate the impact of currency fluctuations, though there is a lack of comprehensive research on the long-term effects of these initiatives on international student mobility (Idachaba, 2014). Additionally, currency volatility can create broader economic instability, affecting job prospects and political conditions in students' home countries, further discouraging them from studying abroad (Ghann & Abraham, 2020). Other critical factors influencing international student flows include GDP, tuition fees, inflation rates, foreign policies, and visa regulations.

2.2.2 Visa Policies, GDP, Tuition Fees and Student Decision-Making

Recent UK visa policy changes, such as stricter regulations on dependents and post-study work, have negatively impacted student mobility, as evidenced by the decrease in net migration of non-EU international students from 328,000 in December 2022 to 247,000 in December 2023 (ONS, 2024). However, the introduction of the Graduate Route visa in 2021 has positively influenced student mobility, with 32% of students opting for this route (MAC, 2024).

The economic health of both home and host countries, indicated by GDP, also plays a crucial role. Students from countries with lower GDPs often seek better opportunities abroad, while strong economies like the UK attract international students due to perceived better job prospects and a higher standard of living. This is supported by data showing increased enrolments from economically growing countries like India and Nigeria (Weber & Van Mol, 2023). Notably, countries with high GDPs, such as the United States (where average tuition fees for international students range from \$26,820 at public universities to \$39,726 at private universities) and Germany (where tuition is often free for international students at public universities), also attract a significant number of international students due to their robust economies and renowned educational institutions.

Tuition fees are a critical factor influencing international student mobility. The UK's average tuition fees for international postgraduate students have risen substantially since the removal of the cap in 2012, now ranging from £12,000 to £25,000 annually for most programs, with specialised courses like MBAs and medical degrees costing up to £63,000 per year (Bennett M, 2024.; Playdon, J, 2018). This is further corroborated by data from the Higher Education Statistics Agency (HESA), which reveals an 18.4% increase in median fees for non-EU postgraduate taught students and a 12.9% increase for non-EU postgraduate research students between 2016/17 and 2021/22 (HESA, 2024). Such high costs can deter prospective students, especially those from lower-income backgrounds, despite the perceived value of a UK degree and potential return on investment (Luo et al., 2023).

Comparatively, tuition fees in the UK are higher than other popular destinations like Germany and France, where public universities often charge nominal fees or none at all (Weber & Van Mol, 2023). In contrast, tuition fees in countries like the US (averaging \$26,820 at public universities and \$39,726 at private universities) and Australia (averaging AUD\$30,840 per year) are also high, creating a complex decision-making process for prospective students who need to weigh the costs against the perceived benefits of studying in these countries.

2.3 University-Level Factors Influencing Enrolment

2.3.1 Financial Aid, Support Services, Recruitment Strategies, Academic and Social Integration Programs

Financial aid has a significant impact on the enrolment of students in universities by directly affecting their capacity to afford higher education. According to research conducted among agriculture undergraduate students in Botswana, the availability of financial aid has a considerable impact on enrolment decisions, with students favouring

institutions that provide substantial financial support (Moakofhi et al., 2019). In the same vein, research conducted on Malaysian private higher education institutions demonstrated a positive correlation between international student enrolment and financial aid, including scholarships and grants (Cynthia & Chong, 2023). The influence of financial aid is not limited to enrolment; it also affects retention and completion rates, as students who receive sufficient financial assistance are more inclined to continue their education (Luo, 2023).

Statistical evidence from the Higher Education Statistics Agency in the UK supports the positive correlation between increased enrolment rates and financial aid. Arrowsmith, J (2023) publication also emphasises the crucial role of financial support in enabling students to access and complete higher education. Their report highlights how implementing maintenance grants and bursaries has led to higher participation rates for students from lower-income backgrounds. This finding is corroborated by HESA data in 2020, which shows that expanding financial aid programs has significantly increased enrolment rates for students from disadvantaged backgrounds. In the 2021/22 academic year, a total of £1.15 billion was awarded in grants and scholarships across the UK. Although this figure doesn't differentiate between domestic and international students, it demonstrates the substantial funding allocated towards financial aid. Furthermore, UK higher education providers received £4.6 billion in 'Other income' during the same period, comprised of; donations and investment income. It's plausible that a portion of this income is directed towards scholarships and grants for international students (HESA, 2023). Additionally, the UK's Tuition Fee Loan system (SLC,2021), which allows students to defer their education costs, has helped maintain enrolment rates despite rising tuition fees.

To improve student experiences and increase retention, it is imperative to provide comprehensive support services, including academic advising, career counselling, and mental health resources. These services establish a supportive learning environment that promotes student success, particularly for international students who receive customised assistance that is tailored to their specific requirements (Luo, 2023). Research on the transition from undergraduate to postgraduate studies among international students in the UK suggests that effective support services are essential for

facilitating this transition and ensuring that students feel integrated and supported (Evans C et al., 2018). Additionally, the reputation of a university is improved by the provision of academic support services, which increases its appeal to prospective students who prioritise a supportive learning environment (Cynthia & Chong, 2023). As emphasised in a systematic evaluation on international student mobility (Luo, 2023), these services also prepare students for their future professions, thereby contributing to long-term success.

Enrolment in universities is substantially determined by effective marketing and recruitment strategies. Prospective students have been demonstrated to be attracted to targeted advertising, campus tours, and engagement with high school counsellors. The study conducted in Botswana underscores the importance of institutional reputation and strategic marketing in shaping students' decisions (Moakofhi et al., 2019). International students are attracted to Malaysian institutions by their diverse program offerings and global reputation (Cynthia & Chong, 2023). Scotland's international education strategy also underscores the significance of comprehensive recruitment initiatives to attract students from around the world (Scottish Government, 2024). Building robust partnerships, personalising student engagement, leveraging digital marketing, and enhancing institutional branding can all contribute to the further development of these strategies.

Enrolment and retention are significantly influenced by academic and social integration programs, which are essential for cultivating a sense of community and belonging. Programs that facilitate academic and social integration, such as orientation programs, peer mentoring, and extracurricular activities, promote a sense of connection among students and their university community. The significance of these programs in improving student satisfaction and retention is emphasised by research on international student mobility (Luo, 2023). The U-curve hypothesis study conducted on international students at a British university further demonstrates that effective integration programs can considerably enhance students' academic and sociocultural adjustment (Xhao & Schartner, 2024).

2.4 Data Science Methods Used to Analyse Student Attrition

2.4.1 Traditional Data Mining Techniques

Autoregressive Integrated Moving Average (ARIMA) models are widely utilised for time-series forecasting in enrolment prediction due to their robust capability in handling temporal data. Dela Cruz et al. (2020) employed the ARIMA(0,2,1) model to forecast student enrolment at Cebu Technological University from 2019-2025, demonstrating an increasing trend in enrolment with statistically significant results, as the model had the lowest Akaike Information Criterion (AIC) value among tested alternatives (Dela Cruz et al., 2020) indicating it was a better model. Similarly, Vanitha and Jayashree (2022) compared ARIMA with Simple Exponential Smoothing (SES) and concluded that while SES performed better for cyclic data, ARIMA was still effective for linear trends (Vanitha & Jayashree, 2022). These studies used historical enrolment data as the primary variable, highlighting ARIMA's effectiveness in capturing long-term trends. However, ARIMA models often face limitations in handling non-linear patterns and require substantial historical data for accurate forecasting, which may not always be available.

Decision Tree algorithms, such as the ID3 algorithm, have been effectively applied to predict student enrolment based on pre-admission criteria. Akinode and Bada (2021) utilised the ID3 Decision Tree algorithm to analyse the impact of high school grades, standardised test scores, and other pre-admission factors on student enrolment at a Nigerian polytechnic. The study reported a 97% accuracy rate, outperforming other algorithms like KNN, SVM, and Naive Bayes (Akinode & Bada, 2021). Similarly, decision trees were used by Shilbayeh and Abonamah (2021) to predict enrolment and attrition at the Abu Dhabi School of Management, yielding high accuracy and providing valuable insights into student behaviour (Shilbayeh & A. Despite their effectiveness, decision trees can be prone to overfitting, particularly with small datasets, and may require pruning techniques to improve generalisability.

The Naive Bayes algorithm, known for its simplicity and efficiency, has been applied in several studies to predict student enrolment. Akinode and Bada (2021) reported an 88% accuracy rate using Naive Bayes, highlighting its potential despite being outperformed by more complex algorithms like decision trees and SVM (Akinode & Bada, 2021). Naive Bayes is particularly useful in scenarios with large datasets and categorical variables, such as student demographics and pre-admission scores. However, its assumption of feature independence often does not hold in real-world data, which can limit its predictive accuracy.

Simple Exponential Smoothing (SES) is another technique used for enrolment forecasting, particularly effective for datasets without clear trends or seasonality. Vanitha and Jayashree found SES more effective than ARIMA in certain educational datasets, providing better performance metrics such as Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) (Vanitha & Jayashree, 2022). SES's simplicity and ability to smooth out noise make it valuable, though it may not capture complex data patterns as effectively as other methods.

Logistic regression has been applied to predict the likelihood of student enrolment based on various predictors. Esquivel and Esquivel developed a logistic regression model to evaluate the probability of admitted students enrolling in a university program, utilising data such as demographic information and academic performance (Esquivel & Esquivel, 2020). The model provided actionable insights for admission strategies. However, it is limited by its inability to capture non-linear relationships.

2.4.2 Machine Learning techniques

Support Vector Machines (SVM) have also been employed in enrolment forecasting due to their robustness in handling high-dimensional data. In the study by Akinode and Bada (2021), SVM was used alongside other algorithms to predict student enrolment,

achieving a commendable 95% accuracy (Akinode & Bada, 2021). The use of SVM in this context demonstrates its effectiveness in dealing with complex, non-linear relationships between variables, such as high school performance and standardised test scores. However, SVMs require careful tuning of hyperparameters and can be computationally intensive, which may limit their applicability in real-time forecasting scenarios.

K-Nearest Neighbours (KNN) algorithm was used by Akinode and Bada to classify student enrolment data, achieving an accuracy of 85% (Akinode & Bada, 2021). While not as accurate as Decision Trees or SVM, KNN provided a straightforward approach to understanding enrolment patterns. KNN's main drawback is its sensitivity to irrelevant features and computational intensity with large datasets.

Ensemble methods, which combine multiple learning algorithms to improve predictive performance, have shown significant promise in enrolment forecasting. Shilbayeh and Abonamah (2021) utilised a boosted regression tree ensemble to predict student enrolment and attrition at ADSM, achieving 89% accuracy, which significantly outperformed single regression trees (Shilbayeh & Abonamah, 2021). Similarly, the study by Vanitha and Jayashree (2022) indicated that ensemble methods, including bagging and boosting, enhanced the predictive power of traditional models (Vanitha & Jayashree, 2022). Despite their effectiveness, ensemble methods can be computationally demanding and complex to implement, requiring careful parameter tuning and validation to avoid overfitting.

Random Forest, an ensemble of decision trees, was used by Al-Shabandar et al (2109) to predict student attrition in online courses, achieving an accuracy of 89%. This technique is robust against overfitting and handles large datasets effectively. However, its complexity and interpretability can be challenging.

Gradient Boosting was used by Al-Shabandar et al. (2019) to predict student dropout, achieving high accuracy. This technique builds models sequentially, with each new model correcting errors of the previous ones. While effective, it is computationally intensive and can be prone to overfitting without proper tuning.

2.4.3 Deep Learning techniques

Artificial Neural Networks (ANN) have been applied in various studies for predicting student enrolment due to their capability to model complex non-linear relationships. For instance, Binu et al. developed an ANN model to predict student enrolment from different states, demonstrating high accuracy in their forecasts (Binu et al., 2015). The study used historical enrolment data, including demographic and academic performance variables, to train the network. The results supported the hypothesis that ANNs can effectively handle large datasets and provide precise predictions, although the model's complexity and need for extensive computational resources were noted as limitations.

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are well-suited for time series forecasting due to their ability to retain information over long sequences. In their study, Fong et al. explored using RNNs and LSTMs for predicting university admissions based on historical enrolment data, achieving significant predictive accuracy (Fong et al., 2009). Variables included previous enrolment figures, academic performance metrics, and demographic data. The effectiveness of these models in capturing temporal dependencies supported the hypothesis that RNNs and LSTMs are highly effective for enrolment forecasting. However, the study noted challenges with model training times and the potential for overfitting, especially with small datasets.

Deep Belief Networks (DBNs) have been employed to model complex hierarchical relationships in student enrolment data. A study by Nakhkoba and Khademi (2016) utilised

DBNs to predict student enrolment, integrating various academic and demographic factors. The DBN outperformed traditional machine learning methods, highlighting its capability to capture intricate patterns in the data. Despite these strengths, the research acknowledged the need for substantial computational power and extensive parameter tuning, which can limit practical applications.

Autoencoders have been used for dimensionality reduction and feature extraction in enrolment forecasting models. Vanitha and Jayashree applied autoencoders to preprocess large educational datasets before applying predictive models (Vanitha & Jayashree, 2022). This technique effectively reduced the data's complexity while preserving critical information, leading to improved model performance. The study highlighted autoencoders' effectiveness in handling high-dimensional data but also noted the complexity involved in training these models and the potential for loss of interpretability.

Hybrid models combining multiple deep learning techniques have shown promise in enrolment forecasting. For example, Fong et al. proposed a hybrid model integrating ANN and decision trees to enhance prediction accuracy (Fong et al., 2009). This approach leveraged the strengths of both methods, resulting in improved performance metrics. However, the study pointed out that hybrid models could be computationally intensive and require careful integration to avoid overfitting and ensure robust predictions.

2.5 Review Summary, Gaps, and Contribution

While Tinto's (1975) student drop-out model observes the demographic composition, and socio-economic factors responsible for fluctuations in retention and drop-out rates, existing student mobility models like the TBP and the push-pull framework often overlook challenges unique to postgraduate students. Aisling found that economic volatility regarding international postgraduate students' home countries was responsible for their

inability to study in Scotland. Factors like differences in currency valuation, visa regulations, GDP, and tuition fees transactions, often determine the mobility of international postgraduate students. For instance, the depreciation of the Nigerian Naira has been linked to a significant drop in the number of Nigerian students in the UK (Mok et al, 2022)

Research acknowledges the importance of financial aid, support services, and recruitment strategies in attracting international students to HEIs (Moakofhi et al., 2019; Arrowsmith, J, 2023), there is a significant gap in understanding these factors' impact specifically on postgraduate students. Furthermore, data science techniques show promise in predicting student enrolment, a more comprehensive approach integrating multiple models could yield more accurate predictions, particularly for postgraduate this current research.

This research aims to address these gaps by investigating the impact of economic factors on postgraduate student enrolment in Scotland, utilising advanced data mining and machine learning techniques to uncover hidden patterns and relationships, ultimately informing more effective recruitment strategies, financial aid programs, and policy interventions for Scottish institutions.

Chapter 3: Methodology

3.1 Research Design

3.1.1 Project Plan

This research, aiming to analyse the relationship between currency fluctuations and international postgraduate student enrolment in Scotland (2014-2022), was structured around an 8–10-week project plan, detailed in **Project Plan.xls**. Table (7.1) provides a roadmap of the key activities undertaken. Due to time constraints, not all tasks, particularly those related to model building and evaluation, were completed. The study analyses historical data on enrolment, visa policy changes, exchange rates, GDP growth, inflation, and other relevant factors to identify patterns, trends, and causal relationships. Findings will provide valuable insights for university administrators, policymakers, and international student recruitment officers seeking to develop effective strategies for attracting and retaining international students.

A quantitative research approach has been adopted for this study. This approach enables the use of statistical techniques to measure and quantify the impact of various factors on student enrolment and to develop predictive models.

3.1.2 Role of Secondary Data Analysis

This study is based majorly on secondary data owing to the availability of legitimate and credible data sources and time limitation on accessing primary information. Thus, data from several sources including, The Higher Education Statistics Agency, government statistical offices and SFC, financial databases such as, Investing UK, World Bank, Statistica, IMF and several other university websites are used. The use of secondary data allows for a comprehensive analysis of historical trends and patterns without the need for primary data collection and most importantly aids predictability.

3.1.3 Research Strategy

While using a quantitative approach to address the aims of this research, the strategy will employ a cross-sectional research design to examine the relationship between the target variable (enrolment numbers “Enrolment”) and other factors which are further grouped into; independent, dependent and categorical variables.

Descriptive statistics is used to summarise the data associated with each variable by looking into means, median, distributions etc. Correlation analysis will be conducted to assess the relationships between enrolment, exchange rates, GDP growth rates, and inflation rates. Regression analysis, including linear, ridge, and lasso regression, will be utilised to model the impact of exchange rate fluctuations, GDP growth rate, inflation rate, and potentially visa-related variables (if included) on enrolment while controlling for other factors like tuition fees. Time series analysis, incorporating ARIMA and LSTM models, will be employed to identify trends and patterns in enrolment over time and to develop predictive models. Additionally, machine learning models such as SVM, Random Forest and XGBoost will be explored to capture non-linear relationships and improve predictive accuracy (Al-Shabandar et al., 2019; Akinbode & Bada 2021; Shilbayeh & Abonamah 2021).

For pre-processing, data analysis and modelling, R will be the primary tool due to its comprehensive statistical capabilities, visualisation options and machine learning tasks due to its efficient libraries and scalability. Excel will be utilised for initial data cleaning and preparation (table 7.1).

Limitations of this study may include reliance on secondary data which may not capture all relevant variables, the potential for omitted variable bias and time constraints.

3.2. Data Sources

All data sources were obtained from valid and credible sources to support data integrity, credibility, and transparency. The table (3.1) below shows all data sources

Data Category	Source	Content	Link
Enrolment Data by country of origin, region and University	Higher Education Statistics Agency (HESA)	- Table 1- HE student enrolments by HE provider 2014/15 to 2021/22 - Table 28 - Non-UK domiciled HE students by HE provider and country of domicile 2014/15 to 2021/22 - Table 11 - HE student enrolments by domicile and region of HE provider 2014/15 to 2021/22	- https://www.hesa.ac.uk/data-and-analysis/students/table-1.csv - https://www.hesa.ac.uk/data-and-analysis/students/table-28.csv - https://www.hesa.ac.uk/data-and-analysis/students/table-11.csv
Tuition Fees	All universities in Scotland, individual website(s). Study Abroad Aide	Range and summary tuition fees across the 18 universities in Scotland from 2014/2015 - 2021/2022	- https://studyabroadaide.com/list-cheap-universities-scotland/ The compiled list is in the Appendix , folder “Project Data”. File “tution_fees_raw.csv”.
Currency Data	Investing.com Oanda	Currency exchange rates for 15 countries (Gambia, Ghana. Kenya. Libya, Nigeria, Tanzania, Uganda, China, India, Japan, Pakistan, Thailand, Kuwait, Oman, Turkey) from 2014-2022	- https://uk.investing.com/currencies/gbp-kes-historical-data - https://fxds-hcc.oanda.com/# The compiled list is in the Appendix folder, “Project Data”, rates_semi_clean.csv”
Economic Indicator Data	World Bank Group Data Statistica.com	GDP and Inflation rates for the 15 countries (Gambia, Ghana. Kenya. Libya, Nigeria, Tanzania, Uganda, China, India, Japan, Pakistan, Thailand, Kuwait, Oman, Turkey) from 2014-2022	- https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG?locations=IN - https://www.statista.com/statistics/256598/global-inflation-rate-compared-to-previous-year/ The compiled file is in the Appendix folder . “ Project

			Data", "gdp_inf_semi_clean.csv"
Visa Changes and Policies data	GOV.UK	Summary of student visa changes, fees and policies from 2014-2024.	https://www.gov.uk/graduate-visa https://www.gov.uk/skilled-worker-visa https://www.gov.uk/health-care-immigration-application/how-much-pay Compiled document is seen in the Appendix, Folder "Project Data", Visa Types .csv

Table: 3.1 Shows data sources, a summary of content and links

3.3 Data Collection and Preparation

The data preparation process involved a series of multiple iterative steps aimed at creating a clean, integrated dataset suitable for analysis. This process (Appendix folder - merged work ugochi.html) included importing various datasets, reshaping them, addressing duplicates, and ensuring data consistency through aggregation and appropriate joins. Below is an overview of the process employed:

Data Importation and Initial Preparation

The data preparation process commenced with the importation of three primary datasets. The merged_train_1.csv dataset provided initial student enrolment figures, while the rates_semi_clean.csv dataset contained historical exchange rates for various countries. A third dataset, merged_1422_result.csv, was incorporated later to facilitate merging and aggregation tasks. To simplify subsequent data manipulation and analysis, all datasets, initially in a wide format with multiple country-specific columns, were reshaped into a long format.

Reshaping and Aggregation

Reshaping the datasets into a unified structure was essential. Using the `pivot_longer()` function, the exchange rate data was transformed from a wide to a long format, consolidating information into key columns: "Country" and "FX_Rate." This restructuring streamlined the integration of exchange rate data with other datasets. Similarly, the enrolment data was reshaped to align country-specific data by year, facilitating aggregation and merging. Duplicate records, identified during this phase, were addressed by grouping data based on key identifiers, ensuring the retention of unique entries.

Data Cleaning and Consistency

To ensure data integrity, rigorous cleaning and consistency checks were implemented. Columns deemed irrelevant to the analysis, such as "Month" and "Date," were removed. Data types were standardised across the merged datasets, particularly for columns used as keys, such as "Fiscal Year" and "Country," to prevent errors during merging. Aggregation was performed on the "Enrolment" and "FX_Rate" columns to ensure that each key combination had a single, summarised value, eliminating redundancy and enhancing the dataset's focus.

Merging and Final Dataset Preparation

The final stage involved merging the cleaned and reshaped datasets. Left joins were employed to preserve all enrolment records, even in instances of missing auxiliary data, such as exchange rates. Additional datasets containing economic indicators, including GDP and inflation, were merged using similar procedures. Key matching was rigorously verified, and inconsistencies in naming conventions were standardised, particularly for fields like "HE.Provider." A final check to remove any remaining duplicates and irrelevant columns resulted in a clean, consistent dataset prepared for analysis.

3.4 Types of Variables and Descriptions

Dependent (Target) Variable

- Enrolment: This variable represents the number of international postgraduate students enrolled in Scottish universities.

Independent Variables

- FX_Rate: This variable encompasses historical exchange rates of students' home currencies against the British Pound.
- Growth.Indicator / Value: Representing the GDP growth rate of the source countries, this variable reflects country's economic health.
- Growth.Indicator: This variable denotes the inflation rate in source countries.
- Tuition_Av: This variable represents the average tuition fees for international postgraduate programs in Scotland.
- Visa.Code: This variable categorises the type of visa obtained by students.
- Visa.Fee.....: This variable captures the cost of obtaining a visa.
- Visa.NHS.Charge.....: This variable represents the NHS surcharge levied as part of visa applications.
- Visa.After.Graduation: This variable indicates the availability of post-graduation visas.
- Visa.Work.Requirements.After.Graduation: This variable outlines work requirements associated with visas.
- Salary.Requirements: This variable denotes the minimum salary thresholds for work visas post-graduation.
- Dependents.Allowed: This variable indicates whether students can bring dependents.
- Visa.Fee..Dependents: This variable captures the costs associated with bringing dependents.
- Visa.NHS.Charge.Depen: This variable represents the NHS surcharge applicable to dependents.
- Fiscal_Year: This variable represents the time periods from 2014-2022 to match with the growth indicators, exchange rates, tuition fees, visa changes etc.

- Academic_Year: This variable represents the start and end of the academic year (2014/2015- 2021/2022)

Categorical Variable

- Country: Treated as a categorical variable, the country of origin may interact with other independent variables (FX_rate, Growth.Indicator).
- HE.Provider: Also considered a categorical variable, represents the 18 different HEIs in Scotland.

3.5 Exploratory Data Analysis

The dataset summary (fig 3.1) spans from 2014 to 2022, listing information related to international postgraduate students across 15 countries, including China, Gambia, Ghana, India, Japan, and Kenya, with each country contributing 560 records (each country is well represented to avoid bias). It covers 18 HE providers, with each listed university having 480 entries. The mode of study is almost an even split between full-time (4,170) and part-time (4,230) studies (fig 3.1). Academic years are consistently represented, with slight decreases in records in the latter years, such as 1,050 entries in 2017 and 2018, and 1,020 in 2019.

Visa information is detailed with multiple visa types, such as "Student Visa," "Tier 4 (General)," and "Tier 4 Pilot," with the highest count being 3,180 for the "Tier 4 Pilot" visa (see fig 3.5). Visa fees and NHS charges vary, with fees ranging from 310 to 348 and NHS charges from 150 to 470. The dataset also highlights post-graduation visa statuses and work requirements, with 3,180 entries for "Tier 2 Only" and 3,060 for "Graduate Route for 2 years." Financial data such as tuition averages range from 12,000 to 23,250, and FX rates show a wide range from 0.388 to 4,971.356. The economic indicators include GDP and inflation metrics, with enrolment figures varying widely, indicating the diverse and complex nature of international postgraduate education trends during this period.

```

## Fiscal_Year      Country      HE.Provider
## 2014 :1080 China : 560 Abertay University : 480
## 2015 :1080 Gambia : 560 Edinburgh Napier University : 480
## 2016 :1080 Ghana : 560 Glasgow Caledonian University : 480
## 2017 :1050 India : 560 Glasgow School of Art : 480
## 2018 :1050 Japan : 560 Heriot-Watt University : 480
## 2019 :1020 Kenya : 560 Queen Margaret University, Edinburgh: 480
## (Other):2040 (Other):5040 (Other) :5520
## Level.Of.Study Mode.Of..Study Academic.Year
## All postgraduate:8400 Full-time:4170 2014/2015:1080
## Part-time:4230 2015/2016:1080
## 2016/2017:1080
## 2017/2018:1050
## 2018/2019:1050
## 2019/2020:1020
## (Other) :2040
## Visa.Code Visa.Fee.... Visa.NHS.Charge....
## Student Visa :2040 310:1080 150:5340
## Tier 4 (General) :2160 322:1080 300:2040
## Tier 4 + Graduate Route:1020 328:2130 470:1020
## Tier 4 Pilot :3180 348:4110
##
##
## Visa.After.Graduation Visa.Work.Requirements.After.Graduation
## Not Applicable:2160 Company Dependent :3180
## Tier 2 Only :3180 Graduate Route for 2 years:3060
## Yes :3060 None :2160
##
##
## Salary.Requirements Dependents.Allowed Visa.Fee..Dependents
## 20800 :3180 Yes:8400 310:1080
## None :2160 322:1080
## Not Applicable:3060 328:2130
## 348:4110
##
##
## Visa.NHS.Charge.Depen Growth.Indicator Enrollment Value
## 150:5340 GDP :4200 Min. : 0.0 Min. : -58.320
## 300:2040 Inflation :4200 1st Qu.: 0.0 1st Qu.: 1.440
## 470:1020 Median : 0.0 Median : 4.120
## Mean : 314.7 Mean : 4.808
## 3rd Qu.: 60.0 3rd Qu.: 6.850
## Max. :97140.0 Max. :153.490
##
## Tuition_Av FX_Rate
## Min. :12000 Min. : 0.388
## 1st Qu.:14500 1st Qu.: 5.645
## Median :16000 Median : 64.833
## Mean :16608 Mean : 585.040
## 3rd Qu.:18500 3rd Qu.: 150.850
## Max. :23250 Max. :4971.356
##

```

Fig: 3.1: Code snippet of the final dataset summary (final_combined_1422) for exploration and analysis

3.5.1 Numerical Variable Distribution

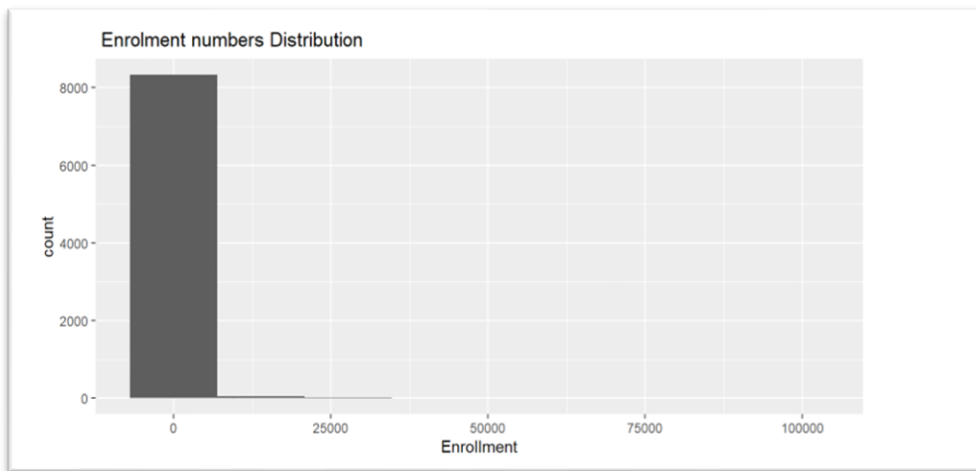


Fig 3.2: Histogram showing the Initial distribution of enrolment numbers in the dataset

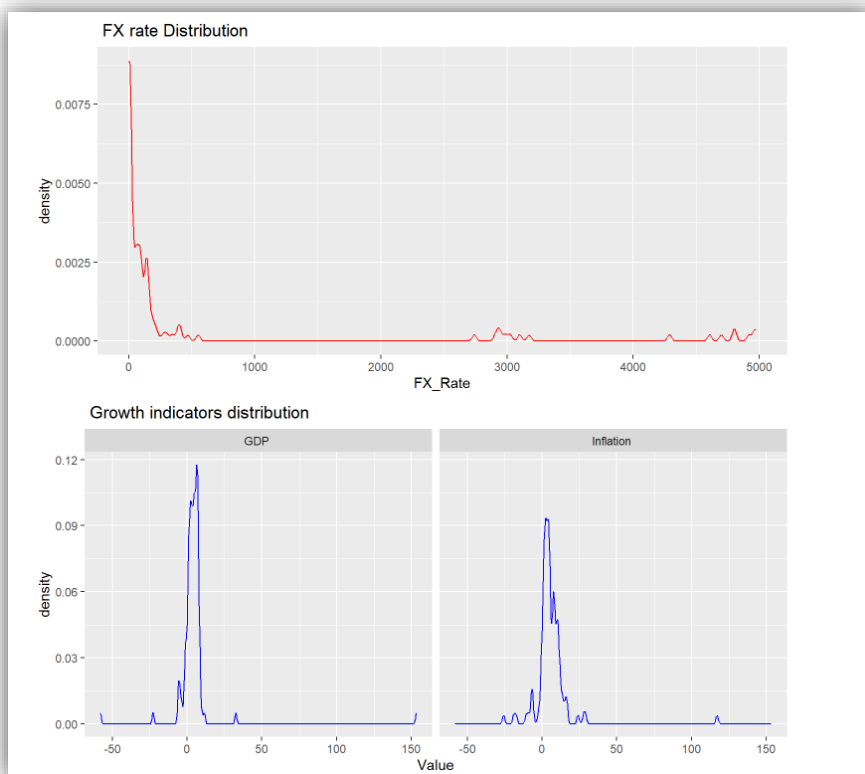


Fig 3.3: Density plots showing the Initial distribution FX rates (in red) and Growth Indicators in blue (GDP rate left and Inflation rate right)

Enrolment numbers as seen in fig (3.2) are largely concentrated at the lower end of the scale, with most data points clustered near zero, indicating that most institutions or countries have relatively low enrolment numbers. The distribution is significantly positively skewed, with a steep decline after the initial peak, and only a few instances where enrolment numbers are significantly higher, suggesting the presence of outliers with much larger enrolments. Similarly, the FX rate distribution (Fig 3.3) is also right skewed, with the same characteristics as enrolments. This distribution implies that while most countries maintain stable, low FX rates, a few exhibit higher, more volatile rates, likely reflecting specific economic conditions.

The growth indicators (fig 3.3) show a concentration of values around zero, with a relatively symmetrical distribution in the case of inflation and some extreme outliers for both variables. The peaks around zero suggest that most countries are experiencing low or moderate GDP growth and inflation, with only a few cases indicating more extreme economic conditions. In contrast, average tuition fees exhibit a normal distribution though still slightly skewed to the right (Fig 3.4). The tuition fees mostly fall within the range of £12,000 to £24,000, reflecting variability in pricing across institutions, likely influenced by factors such as institutional reputation, geographic location, and course offerings. This spread indicates a broader range of tuition fee structures compared to the other variables in the dataset

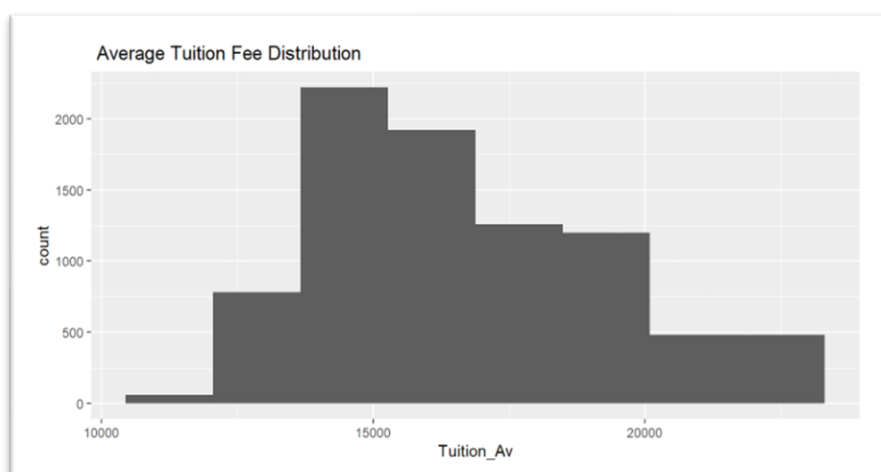


Fig 3.4: Histogram showing the Initial distribution average tuition fees across the dataset

3.5.2 Categorical Variable Distribution

As mentioned previously the distribution of Visa Codes shows that the "Tier 4 Pilot" category has the highest frequency (Fig 3.5). This suggests that the Tier 4 Pilot visa was in operation for a longer period compared to other visa types. In terms of Modes of Study (Fig 3.1), there is an almost even split between full-time with a slight preference for part-time study, indicating that both modes are similarly popular among students. The distribution of Higher Education (HE) Providers is mostly uniform, except for SRUC, which has a noticeably lower count (Fig 3.6), suggesting it might be a smaller institution. The class distribution of Growth Indicators, specifically GDP and Inflation, is balanced, with 4,200 records each (Fig 3.1), allowing for an equitable comparison between these economic factors. Additionally, the country distribution is evenly spread, with 560 (Fig 3.1) records for each of the listed countries, ensuring that analyses are not biased by the overrepresentation of any country.

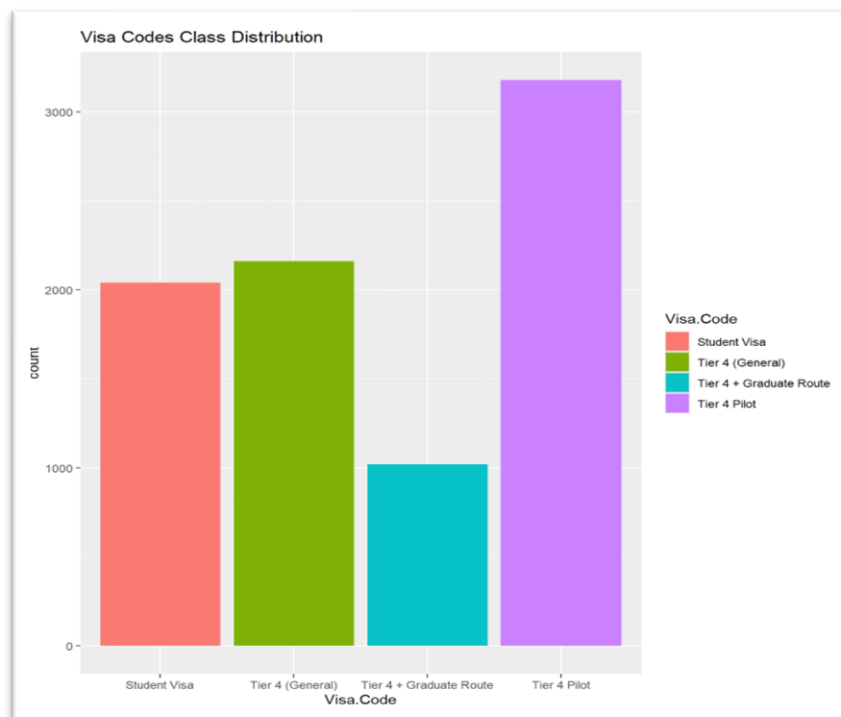


Fig 3.5: Histogram of the initial distribution of the Visa codes in the dataset

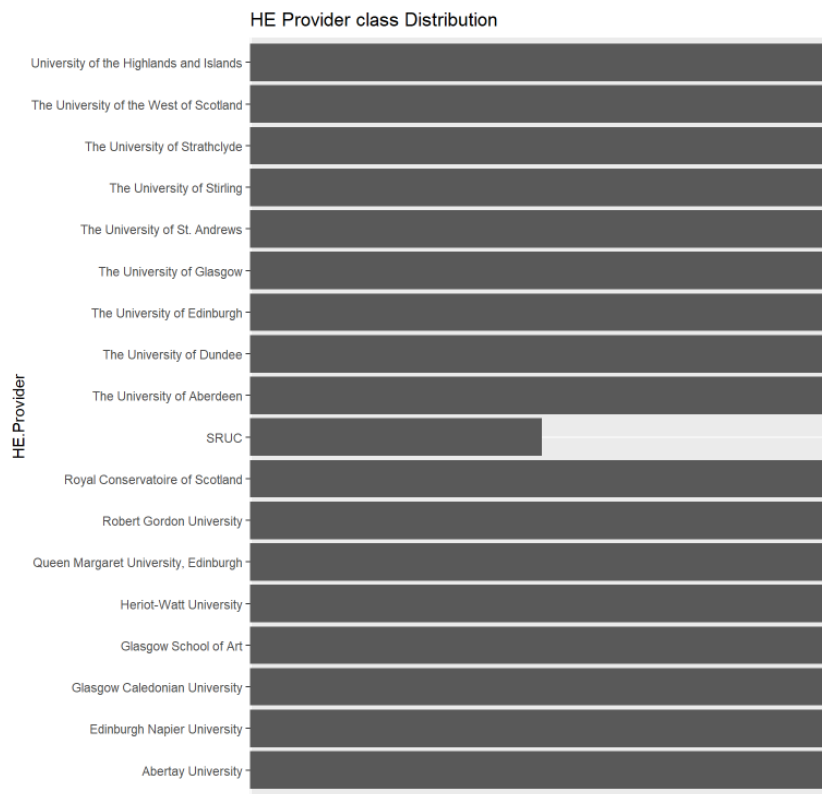


Fig 3.6: Histogram of the initial distribution of the Universities in the dataset

3.5.3 Paired Distribution (Numerical & Categorical Variables)

The analysis of the 4 plots (fig 3.7-fig 3.9) provides observations on the cumulative FX rates and GDP values for various countries relative to the British pound over the fiscal years within the dataset. The FX rate plots (fig 3.7) reveal that Uganda and Tanzania have the highest cumulative FX rates, indicating that their currencies have consistently depreciated more significantly relative to the British pound compared to other countries. The countries could be possible outliers in the FX rate distribution. Conversely, countries like China, Japan, and Ghana have maintained lower cumulative FX rates, suggesting more stable currencies over the same period.

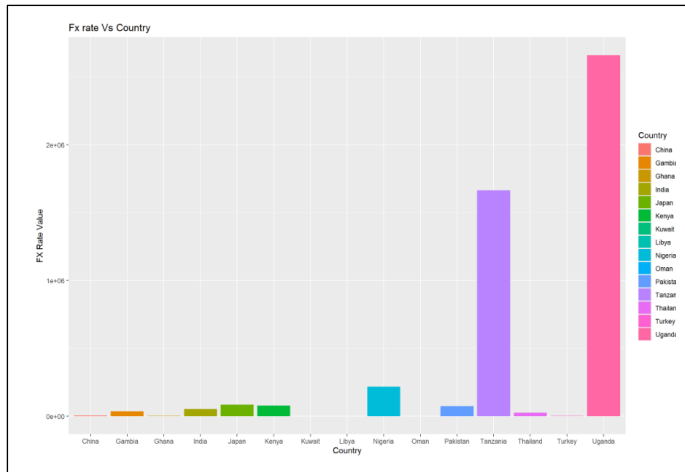


Fig 3.7: Shows FX rate by country

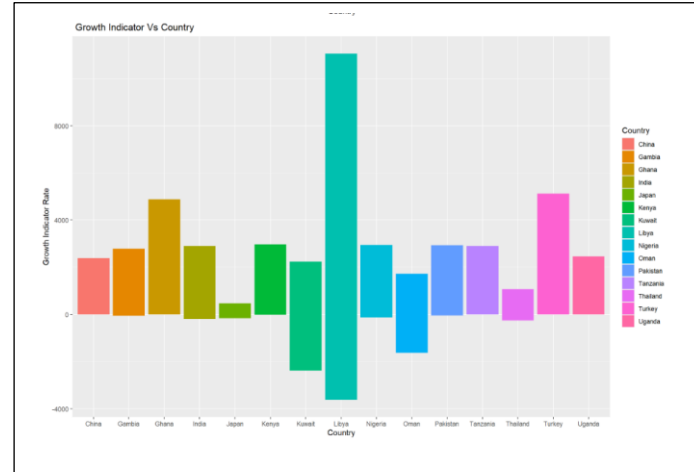


Fig 3.8: Growth Indicators by country

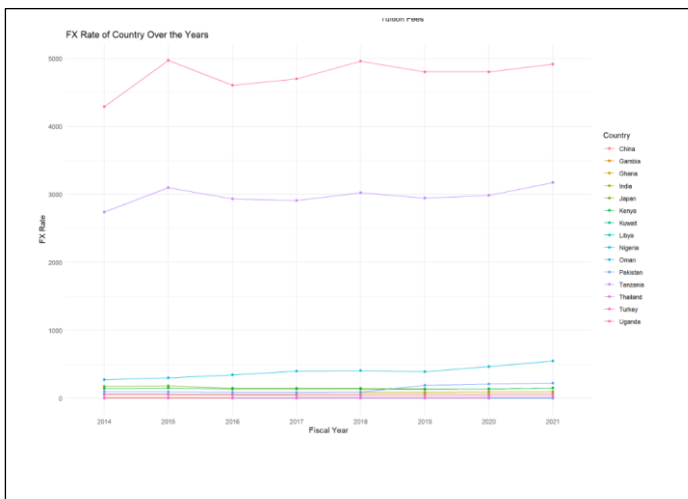


Fig 3.9: Shows yearly FX rate by country indicators by country

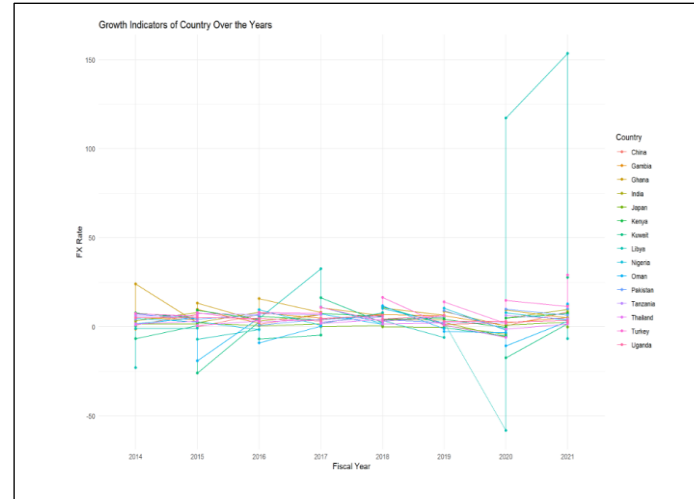


Fig 3.10: Shows yearly GDP

From the plots (Fig 3.8 and 3.10), Libya is observed as an outlier which means there is an extremely high GDP value demonstrating a significant economic performance as compared to the British pound. Nigeria and Turkey are observed as outliers based on their extremely low or in this case, a negative value of the cumulative GDP contrary to the GDP patterns in most other countries. As shown in fig 3.10, suggest that Libya has experienced much stronger economic growth, while Nigeria and Turkey have faced more severe economic challenges.

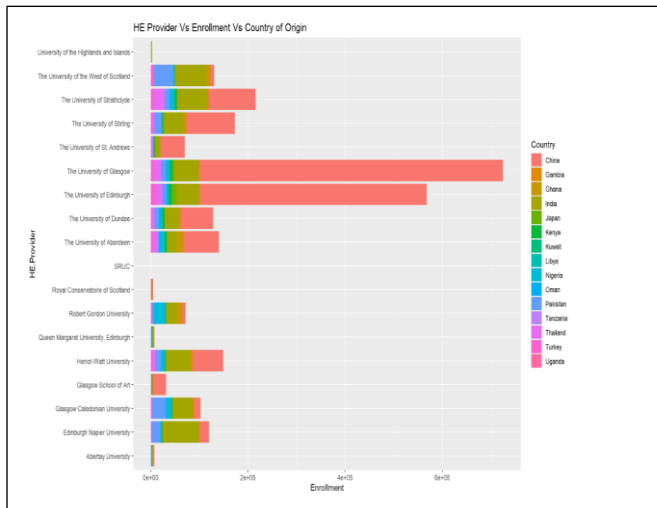


Fig 3.11: University by Enrolment & Country

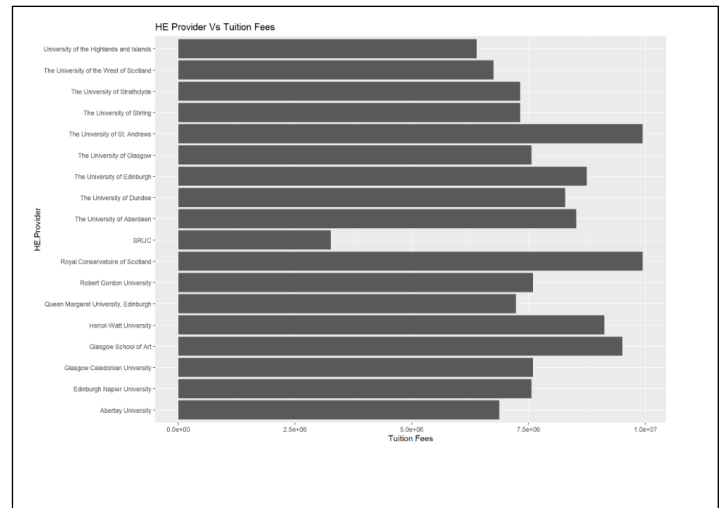


Fig 3.12: HE University by Tuition Fees

The University of Glasgow and the University of Edinburgh as seen in fig (3.11) stand out with the highest enrolment figures, particularly from China, which dominates the enrolment numbers in these institutions. This could also justify the high enrolment numbers seen in fig (3.2) showing a positively skewed distribution. Other HE providers, such as the University of St. Andrews and the University of Strathclyde, also show considerable enrolment, but with more balanced distribution across different countries. SRUC is observed to be an outlier on the lower end, also observed in fig (3.6), with minimal enrolment figures across all countries, indicating either a smaller student body or a specialised focus (agriculture) that does not attract large numbers of international students.

There is a clear correlation between enrolment figures and cumulative tuition fees, with the University of Glasgow and the University of Edinburgh showing the highest tuition fees (Fig 3.12). Most HE providers fall within a similar range for tuition fees, with SRUC being a notable outlier with significantly lower revenue (Fig 3.12). The plot emphasises consistency in tuition fee structures across institutions, with only a few universities standing out due to their higher total revenue. Mid-tier institutions such as Heriot-Watt University, the University of St. Andrews, and the University of Strathclyde also demonstrate substantial tuition revenue. In contrast, SRUC and the University of the

Highlands and Islands have considerably lower total tuition fees, indicating either lower enrolment (Fig 3.11), lower tuition rates, or a combination of both (Fig 3.12).

3.5.4 Outliers

From previous exploration in this section a presence of outliers has been observed in the following variables “Enrolment”, “Value” (which represents GDP and Inflation rates also known as “Growth. Indicator” in the dataset), “FX_Rate and “Tuition_Av”. To detect the outliers a boxplot (Fig 3.13) was used to visualise it, inter quartile range (IQR) (Fig 3.14) was applied to the variables to either remove them or clip them.

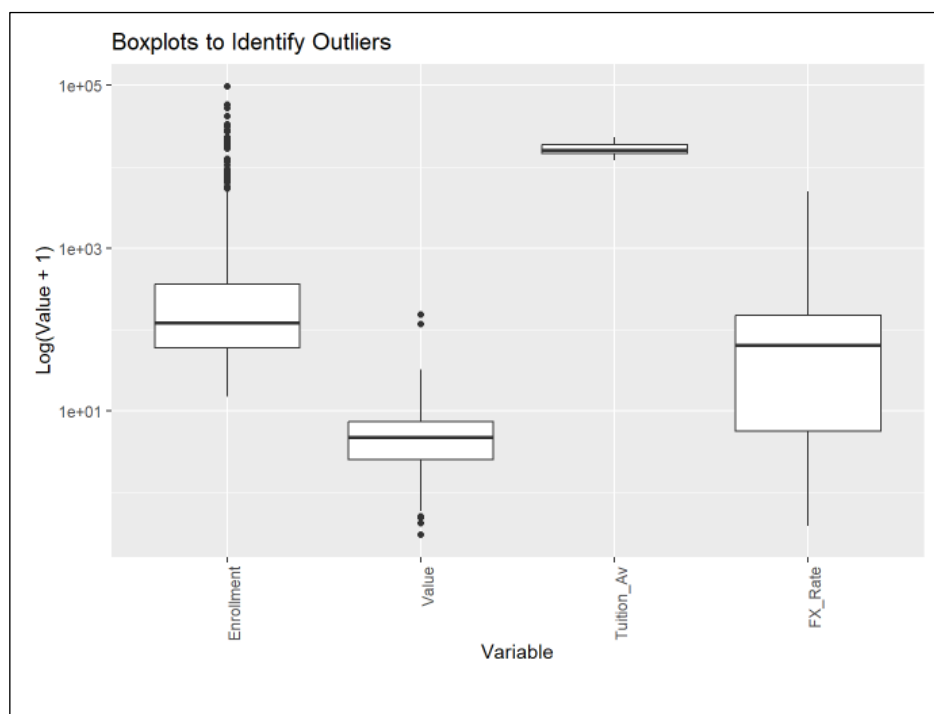


Fig 3.13: Boxplot showing the outliers

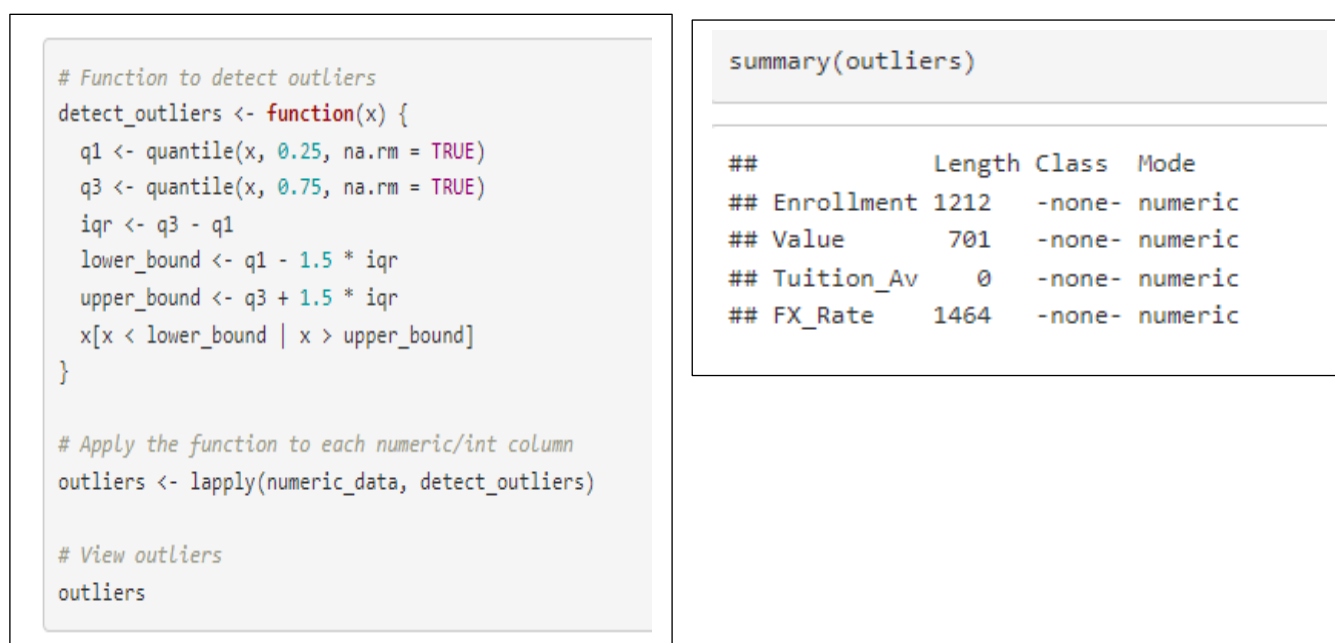


Fig 3.14: Code snippet of IQR applied to outlier and Outlier summary

The boxplots (Fig 3.13) show distribution and variability “Enrolment”, “Value”, “Tuition_Av”, and “FX_Rate”. “Enrolment” reveals considerable variability, with a wide interquartile range (IQR) and a median positioned toward the lower end, indicating that most institutions have modest student numbers. However, the presence of numerous outliers, with values reaching up to 10^5 , highlights that a few institutions enrol significantly more students than others. These outliers likely represent large universities (University of Glasgow or Edinburgh) as seen in fig (3.11) contributing to a skewed distribution where a small number of institutions dominate in student numbers. The “Value” boxplot, which represents economic indicators such as GDP or inflation, shows a more compact IQR, indicating less variability. However, the presence of both high and low outliers suggests substantial deviations, reflecting regions with either strong or weak economic performance.

The “Tuition_Av” boxplot displays a narrow IQR, suggesting that most institutions have similar tuition fee structures, with the median positioned near the top, indicating generally high tuition fees. The relative absence of outliers (Fig 3.14) implies a degree of standardisation in tuition fees, with only a few institutions deviating significantly, likely due to factors like regional economic conditions or institutional prestige.

In contrast, the FX_Rate boxplot shows significant variability, with a wide IQR and a median skewed toward the lower end, suggesting that most currencies maintain relatively low exchange rates. The presence of outliers on both ends indicates notable deviations, with higher outliers reflecting weaker currencies and lower outliers representing stronger currencies.

3.5.5 Outlier Treatment

After application of IQR to treat the outlier two methods were applied only 3 variables (Enrolment, Value and FX_Rate) in the dataset. The methods (fig 3.15) where to remove the outliers completely or to clip them at 95th percentile (retaining all data points while reducing the influence if extreme values. More on this can be seen in the R markdown file (merged_ugochi_work.Rmd) in the Appendix folder.



Fig 3.15: Code snippet of process to remove outliers (left) and clipping at 95% (right)

The no outlier method was chosen to carry out further analysis for this project. When the outliers were removed compared to when clipped the variables (Fig 3:16a-c) became more normalised and less skewed. This allowed for a better understanding of the relationships between variables with any distortion influenced by extreme values.

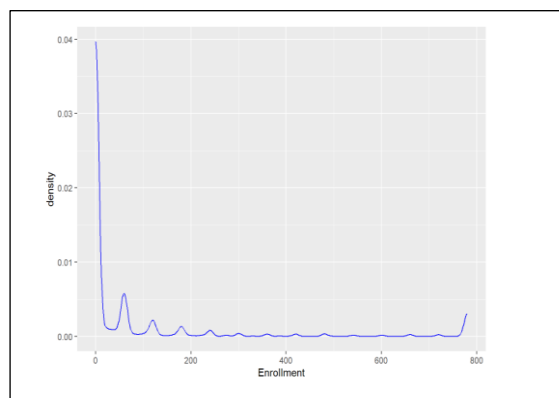
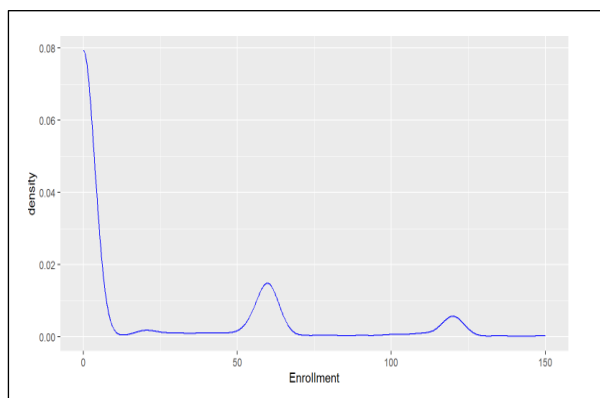


Fig 3.16a: Enrolment density plot distribution No Outlier (left), Clipped (right)

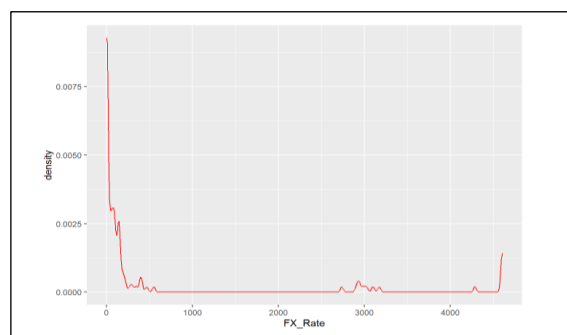
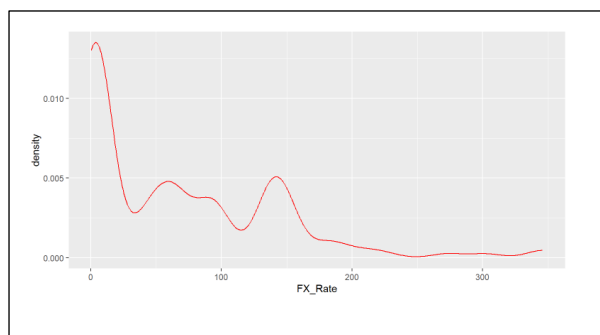


Fig 3.16b: FX_Rate density plot distribution No Outlier (left), Clipped (right)

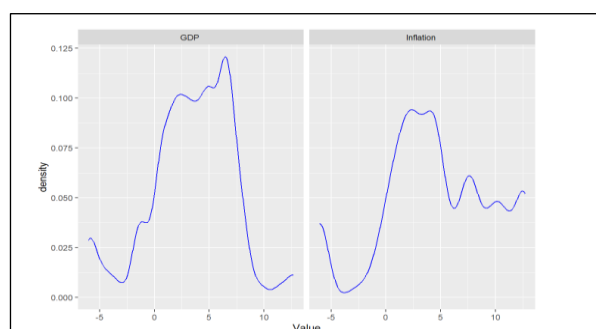
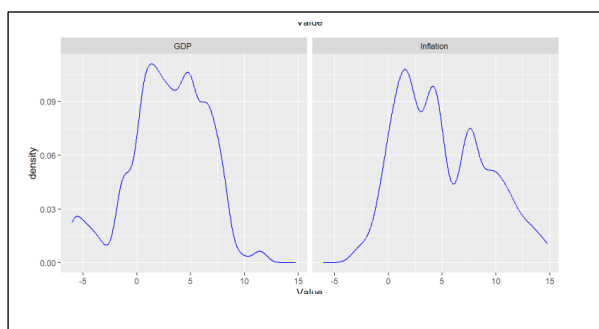


Fig 3.16c: Value density plot distribution No Outlier (left), Clipped (right)

In the “Enrolment” plot (Fig 3.16a), the no-outlier dataset shows a broader distribution with a gradual decline, capturing a wider range of enrolment figures, while the clipped dataset shows a sharper drop, indicating a more constrained view. The “FX_Rate” plot (Fig 3.16b) the no-outlier data displays a complex distribution with multiple peaks, reflecting diverse currency strengths, whereas the clipped data concentrates values near the lower end, missing variability among higher “FX_Rates”. Similarly, “Value” plot (Fig 3.16c) the no-outlier dataset shows a broad distribution with distinct peaks, indicating common economic conditions, whereas the clipped version truncates extremes, reducing the spectrum of economic variability captured. While the clipped datasets provide a streamlined view by removing outliers, they oversimplify the distributions and may omit critical data. In contrast, the no-outlier datasets preserve more natural variability, offering a fuller picture of underlying patterns and making them more suitable for analyses.

3.5.6 Correlation

Pearsons’s correlation was carried out to see the level of significance in the relationship between numerical variables and the target variable (enrolment), see Appendix 2). “FX_Rate” and “GDP” Value show very weak positive correlations with “Enrolment”, with coefficients of $r=0.050$ and $r=0.036$, respectively, both statistically significant (p-values of 0.0003577 and 0.009811). This suggests a slight positive relationship between these variables and enrolment numbers. Conversely, “Visa Fees” exhibit a very weak negative correlation with “Enrolment” ($r=-0.035$, p-value = 0.01185), indicating that higher visa fees may slightly reduce enrolment. The correlation between “Tuition_Av” and “Enrolment” is very weak and not statistically significant ($r=0.023$, p-value = 0.1035), suggesting no meaningful relationship. While some correlations are statistically significant, they appear weak, indicating that these variables may have minimal impact on enrolment. This leads to further investigation of the variables and relationships to identify the strongest predictors using linear regression and feature engineering.

3.5.7 One-Hot Encoding

The categorical variables were converted to binary codes for data anonymisation, evidence of this can be seen in the R markdown file, named “merged_ugochi_work.Rmd”. For correlation was done to investigate relationships between other variables (Fig 3.17) within the dataset and enrolment numbers.

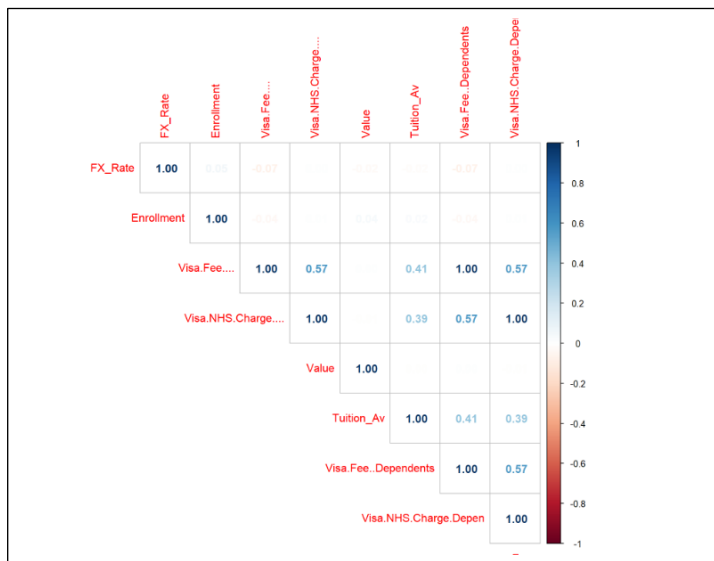


Fig 3.17: Correlation Matric of other numerical variables and enrolment numbers.

The correlation matrix supports the previous findings, showing that FX_Rate, Visa Fees, and Tuition_Av have weak correlations with Enrolment, confirming their minimal direct impact on student numbers. Specifically, FX_Rate and Enrolment have a weak correlation of $r=0.05$, and Visa Fees show a slightly stronger, but still weak, negative correlation with Enrolment ($r=-0.07$). Stronger correlations are observed among financial variables themselves, such as Visa Fees and NHS Charges ($r=0.57$) and between Tuition_Av and Value (Growth indicator) ($r=0.39$), indicating that these costs tend to rise together. Overall, the matrix reinforces that while financial factors are interrelated, their influence on enrolment is limited.

A multivariate correlation was also done to explore further, and the results gave similar outputs. The analysis showed that (see merged_ugochi_work.Rmd) the top 10 factors correlated with enrolment include average tuition fees ($r=0.0605$) and FX rates ($r=0.051$),

both displaying weak positive correlations, consistent with previous findings that these financial variables have a minimal impact on enrolment. Additionally, visa-related factors such as the Tier 4 General visa ($r=0.0332$) and visa fees ($r=-0.0161$) show weak correlations, further supporting the notion that while these factors influence enrolment, their impact is limited. Overall, the results reinforce the earlier conclusion that financial and visa-related factors play a minor role in driving enrolment numbers.

3.5.7 Linear regression

```
##
## Call:
## lm(formula = enrollment ~ fx_rate + visa_fee____ + tuition_av +
##     value, data = final_combined_1422_encoded)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -29.64  -20.17  -17.90   11.50  135.09
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  11.7725730  3.2663857   3.604 0.000316 ***
## fx_rate       0.0230368  0.0067163   3.430 0.000608 ***
## visa_fee____ -1.6934573  0.5196920  -3.259 0.001127 **
## tuition_av    0.0006023  0.0002083   2.892 0.003845 **
## value        0.3373782  0.1256234   2.686 0.007263 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 35.59 on 5070 degrees of freedom
## Multiple R-squared:  0.006561, Adjusted R-squared:  0.005777
## F-statistic: 8.371 on 4 and 5070 DF, p-value: 9.99e-07
```

```
##              IncNodePurity
## fx_rate          523166.0
## visa_fee____     108214.1
## tuition_av       635772.5
## value           337908.2
```

Fig 3.18: linear regression model output

The linear regression report (Fig 3.18) supports previous findings that FX_Rate and Visa_Fee are statistically significant factors influencing enrolment, with FX_Rate showing a small positive effect (coefficient=0.0203, $p=0.0006p$) and Visa_Fee showing a small negative effect (coefficient=-1.694, $p=0.0112$). However, the model's low R-squared value (0.0056) indicates that these variables, along with Tuition_Av and Value, explain only a small fraction of the variability in enrolment. This result contradicts the expectation

that these factors would have a more substantial impact, as suggested by their weak correlations in the earlier analysis. Overall, while the variables are statistically significant, their practical influence on enrolment is minimal, highlighting the need for additional factors or feature engineering to better explain enrolment trends.

3.6 Statistical Tests and Non-Model Based Approaches

There was a focus on using non-model-based approaches on in this project due to time constraints and technical difficulties in testing and evaluation of the trained model, this will be discussed in chapter 5 under limitations.

Correlation (Pearson and Spearman) and Granger Tests

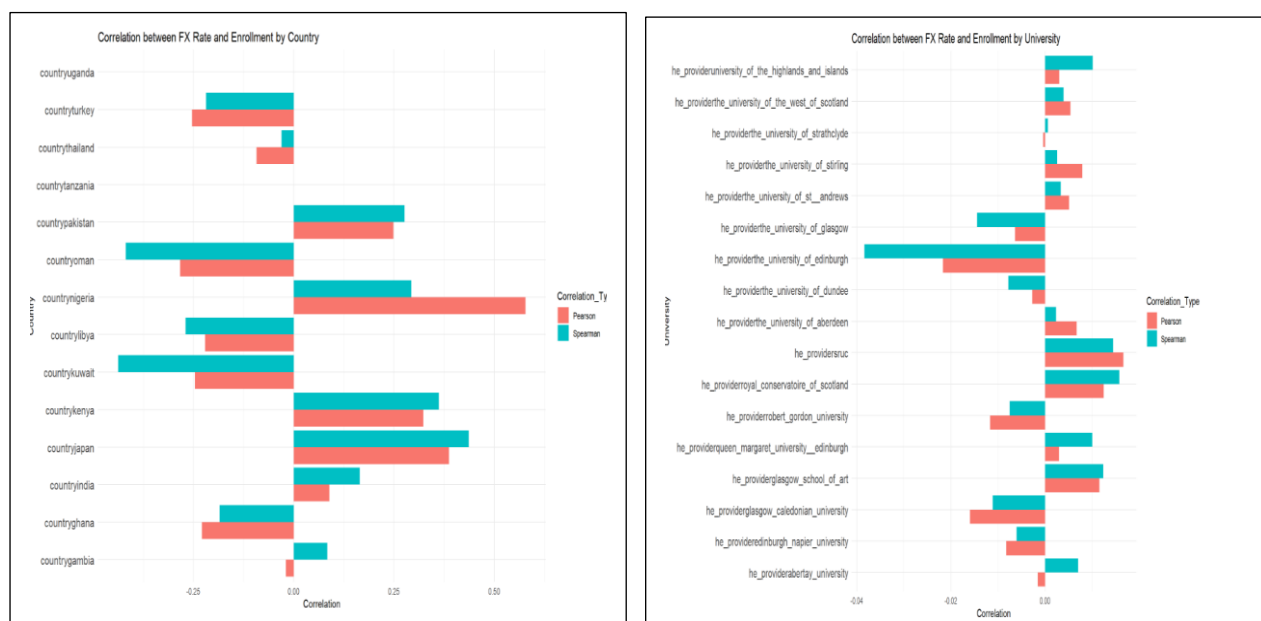


Fig 3.19: Correlation plot Fx rate, by Country Enrolment(left), Fx Rate by University Enrolment (right)

The analysis reveals varying correlations between FX rates and enrolment across countries and universities (Fig 3.19). For instance, Nigeria shows a strong positive correlation, with Pearson's $r=0.50$ and Spearman's $r=0.45$, indicating that as FX rates rise,

enrolment increases. At the university level, the University of Glasgow exhibits a stronger correlation, with Pearson's $r=0.35$ and Spearman's, suggesting that changes in FX rates are more likely to influence enrolment there. However, the Granger causality test (Fig 3.20), with an F-statistic of 2.5945 and a p-value of 0.1073, indicates that FX rates alone do not significantly predict future enrolment changes.

Granger Causality Test: Test if changes in exchange rates can predict changes in enrollment

```
library(lmtest)
granger_test <- grangertest(enrollment ~ fx_rate, order = 1, data = final_combined_1422_encoded)
print(granger_test)
```

```
## Granger causality test
##
## Model 1: enrollment ~ Lags(enrollment, 1:1) + Lags(fx_rate, 1:1)
## Model 2: enrollment ~ Lags(enrollment, 1:1)
##   Res.Df Df       F Pr(>F)
## 1    5071
## 2    5072 -1 2.5945 0.1073
```

Fig 3.20: Code snippet of Granger Causality test

```
##           Visa_Code Pearson_FX_Rate Spearman_FX_Rate
## 1   visa_codetier_4_(general)    0.06576938    0.06627634
## 2 visa_codetier_4_+graduate_route -0.06212150   -0.08202093
## 3   visa_codetier_4_pilot    -0.04516495   -0.07988335
##   Pearson_Enrollment Spearman_Enrollment Pearson_Tuition_Fees
## 1    0.041595242    0.033238533    -0.33674166
## 2    0.002056478    0.003522853    0.13422570
## 3   -0.042874606   -0.041936748   -0.07563295
##   Spearman_Tuition_Fees Pearson_Value Spearman_Value Pearson_Visa_Fee
## 1   -0.35504489    0.002312744   -0.00314082   -0.8885881
## 2    0.14374743   -0.112704533   -0.12306563    0.3307380
## 3   -0.05399142    0.119818769    0.13287453    0.1937006
##   Spearman_Visa_Fee
## 1   -0.81545495
## 2    0.37228890
## 3    0.05228247
```

Fig 3.21: Correlation Between Visa Types, FX Rates, and Financial Variables Impacting International Student Enrolment

Visa Codes exhibit varied relationships with FX rates, enrolment, and tuition fees (Fig 3.21). The **Tier 4 General Visa** is slightly positively correlated with FX rates ($r=0.066$) but negatively correlated with tuition fees ($r=-0.355$) and visa fees ($r=-0.815$), suggesting that higher costs reduced its applicability. The **Tier 4 + Graduate Route Visa** is negatively correlated with FX rates ($r=-0.082$) and positively correlated with economic indicators

($r=0.1437$), indicating that better economic conditions depicted and increase. The **Tier 4 Pilot Visa** also shows a negative correlation with FX rates ($r=-0.045$). These findings align with previous results, which showed that higher financial costs generally decrease visa applications and enrolment, reinforcing the complex interaction between economic factors and international student behaviour.

3.7 Predictive Insights

Predictive insights were implored to hint predictive relationships by attempting to forecast enrolment based on currency fluctuations.

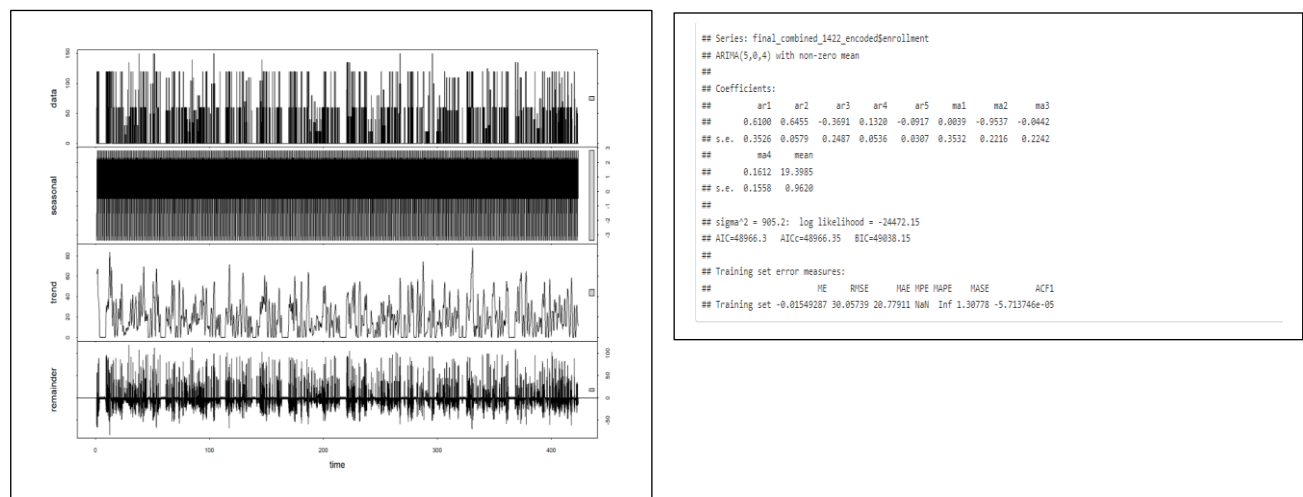


Fig 3.22: ARIMA Model Output (right), Decomposition of Time Series (left)

The ARIMA(5,0,4) model (fig 3.22) applied to the enrolment data shows a mix of significant and less significant AR and MA coefficients, with a log likelihood of -24472.15 and an AIC of 48966.3, suggesting a moderate fit. The model's accuracy, indicated by RMSE (30.05739) and MAE (20.77911), is reasonable, though the high residual error suggests potential issues with noise or model complexity. The time series decomposition further reveals strong seasonal patterns, fluctuating trends, and significant noise in the data, highlighting the challenges in modelling and predicting enrolment trends due to the pronounced variability and unexplained variance.

3.8 Sensitivity and Cluster Analysis

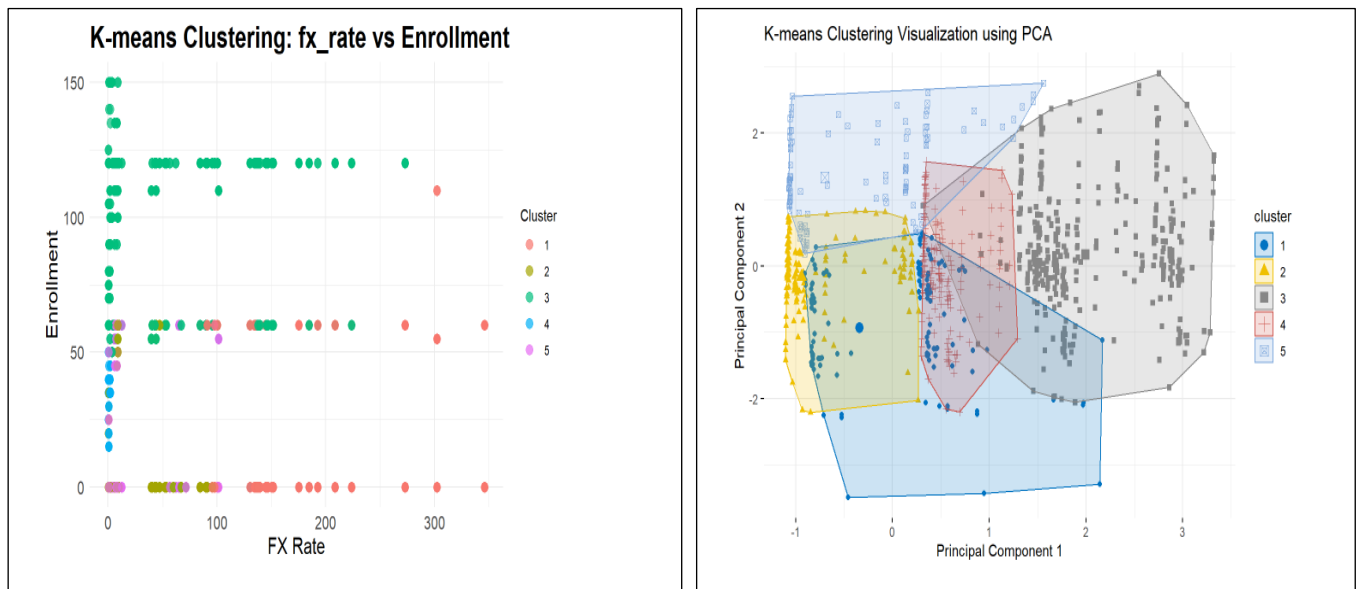


Fig 3.23: Cluster Analysis using Kmeans (left) and PCA (right)

The cluster analysis (Fig 3.23), using PCA and K-means, reveals distinct groupings in the data based on FX rate, enrolment, and mode of study. The first two principal components (PC1 and PC2) capture 55.87% of the variance, with PC1 alone explaining 30.31%. This effective dimensionality reduction allows for clear visualisation of the clusters. The loadings show that enrolment (0.71) and “mode_of_study “(part-time) (-0.69) are the most influential in PC1, while FX_rate (0.63) dominates PC2. The K-means clustering indicates that “FX rate” and enrolment are key factors distinguishing the clusters, confirming that these variables significantly influence the data's structure. Additionally, the sensitivity of students from different countries to currency fluctuations was analysed, revealing that Nigeria showed the highest sensitivity, with a Pearson correlation of 0.5782, while Kuwait exhibited a negative sensitivity, with a Pearson correlation of -0.2465 as seen in fig (3.19). This analysis highlights how financial and enrolment characteristics, combined with country-specific sensitivities, influence the natural grouping within the dataset.

3.9 Feature Engineering Attempt

In this feature engineering attempt, rolling averages for FX rates and economic indicators were calculated to smooth out fluctuations and capture long-term trends, and interaction terms were created to explore combined effects on enrolment. The Random Forest model identified rolling averages (%IncMSE: 62.58, IncNodePurity: 3,119,284.000) and FX rates (%IncMSE: 36.57, IncNodePurity: 138,153.310) as the most influential predictors. This contradicts earlier analyses where tuition fees and visa factors were considered significant, but it complements those findings by highlighting the complex interplay of economic factors. The detailed output can be located at towards the end of the markdown file in the appendix folder.

Despite the model explaining 93.19% of the variance, errors in testing, such as the RMSE and lower importance of some features, suggest room for improvement. These discrepancies indicate the need for future work, focusing on refining the feature set and possibly integrating more nuanced economic indicators to enhance predictive accuracy.

3.10 Legal, Ethical, Social & Professional Issues

3.10.1 Legal Issues

Using only secondary data required strict compliance with GDPR and data protection laws. Data was anonymised using one-hot encoding to prevent re-identification risks. The project adhered to legal requirements by ensuring all data processing aligned with the original collection terms. International data transfers were conducted in compliance with GDPR, verifying safeguards like Standard Contractual Clauses (SCCs). Data retention policies were implemented to store data only as long as necessary, and all third-party data sources, including HESA, World Bank, and Oanda, were properly cited to respect intellectual property rights.

3.10.2 Ethical Issues

To ensure ethical analysis and prevent the reinforcement of stereotypes, the project prioritised fairness and transparency. The analysis process ensured equal representation of variable classes, avoiding bias towards specific universities or countries. Furthermore, the findings were presented with careful consideration of their limitations, avoiding overgeneralisations. This approach aimed to prevent stakeholders from drawing inaccurate or discriminatory conclusions based on the data, such as targeting students from wealthier countries while neglecting broader considerations like cultural diversity and long-term economic trends.

3.10.3 Social Issues

The project acknowledged the potential social ramifications of its findings, particularly concerning equity and accessibility in international education. While the analysis examined the influence of economic factors like currency fluctuations, the project emphasised the importance of promoting diversity and inclusion. Recommendations aimed to avoid exacerbating existing inequalities, such as prioritising students from wealthier nations or inadvertently causing financial hardship for students from specific regions due to tuition adjustments. The project consistently prioritised equitable access to education and considered the broader social impact of its findings.

3.10.4 Professional Issues

Professional responsibility and accountability were paramount throughout the project. Rigorous analysis, including meticulous data checking and sound methodologies, ensured the accuracy and objectivity of the findings. This approach aimed to prevent misinterpretations that could lead to detrimental decisions by educational institutions, such as unjustified tuition increases or misinformed recruitment strategies. Furthermore, the project prioritised clear and precise communication with stakeholders, including universities and policymakers, to ensure that the findings were understood,

and their limitations were recognised, facilitating well-informed decisions based on accurate data interpretation.

3.11. Risk Mitigation

To mitigate potential risks, the project implemented data anonymisation, proper citation practices, and the use of historical secondary data to minimise sensitivity concerns. Equal representation of variable classes and removal of outliers ensured unbiased analysis. Clear communication and balanced analysis were maintained to promote ethical and socially responsible recommendations. Despite these efforts, risks remain related to stakeholder interpretation and the application of findings, emphasising the need for ongoing dialogue and careful consideration in decision-making.

Chapter 4: Discussion

4.1 Correlation Between Currency Fluctuations, Economic Indicators and International Student Enrolment

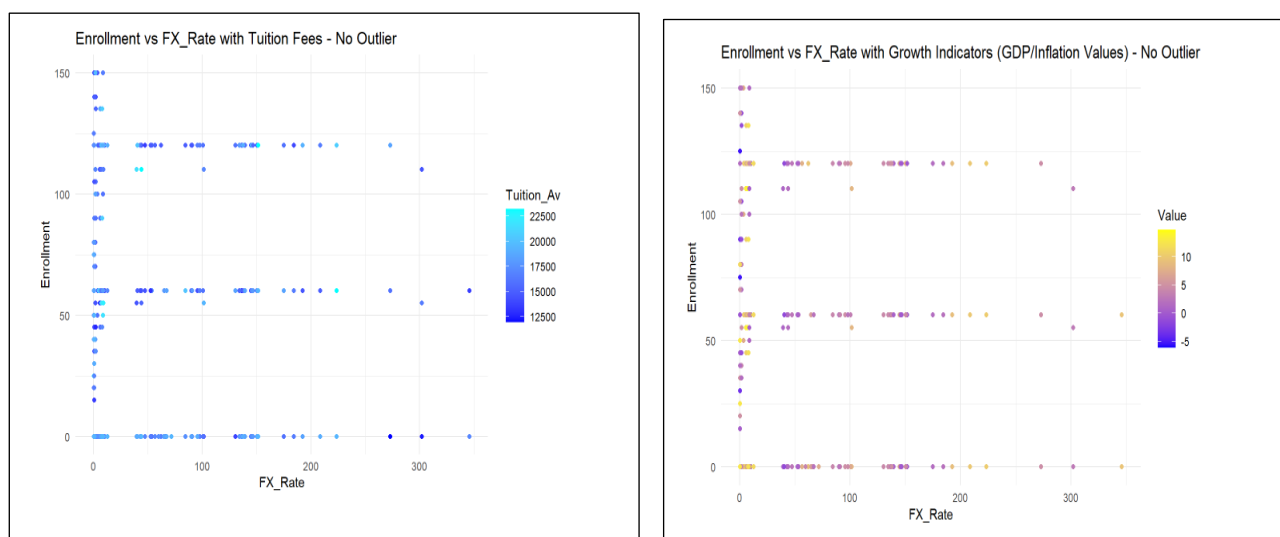


Fig 4.1: Shows impact of FX Rates, Tuition Fees, and Economic Growth Indicators on International Student Enrolment

The analysis of international student enrolment in Scottish universities highlights the significant impact of currency fluctuations, economic indicators, and tuition fees on enrolment numbers. As illustrated in Figure 4.1, there is a clear inverse relationship between FX rates and enrolment, particularly for countries with high currency volatility like Nigeria and Turkey. Higher FX rates correlate with lower enrolment, indicating that currency depreciation against the British Pound increases the financial burden on students, leading to reduced enrolment (Pearson correlation of 0.578 for Nigeria). This is supported by findings from a recent study by Ekugo (2024), which reported that the devaluation of the Nigerian Naira forced some students to discontinue their studies abroad due to increased financial strain.

Economic indicators, such as GDP growth and inflation, also influence enrolment trends. Countries (China and India) with stable economic conditions, as depicted by warmer

colours in Figure 4.1, generally show higher and more consistent enrolment figures. This suggests that students from economically stable countries are more capable of affording international education, resulting in more stable enrolment numbers irrespective of the tuition fee cost. Especially the Chinese who attend the most expensive and top Universities in Scotland (University of Glasgow and Glasgow).

Tuition fees further exacerbate the impact of FX rate fluctuations. Figure 4.1 demonstrates that higher tuition fees are associated with a stronger correlation between FX rates and reduced enrolment. This effect is particularly pronounced for students from countries with weaker currencies, where the financial burden of high tuition fees combined with unfavourable exchange rates significantly lowers enrolment numbers.

These findings align with the work of Chevalier (2022), who emphasises the importance of economic stability and affordability in attracting international students. Similarly, Grebennikov and Shah (2012) argue that understanding these dynamics is crucial for improving student retention and designing effective recruitment strategies. The combined effect of FX rates, economic conditions, and tuition fees underscores the need for targeted strategies by Scottish universities to attract and retain international students effectively.

4.2 Enrolment Trends and Patterns in Scottish Universities

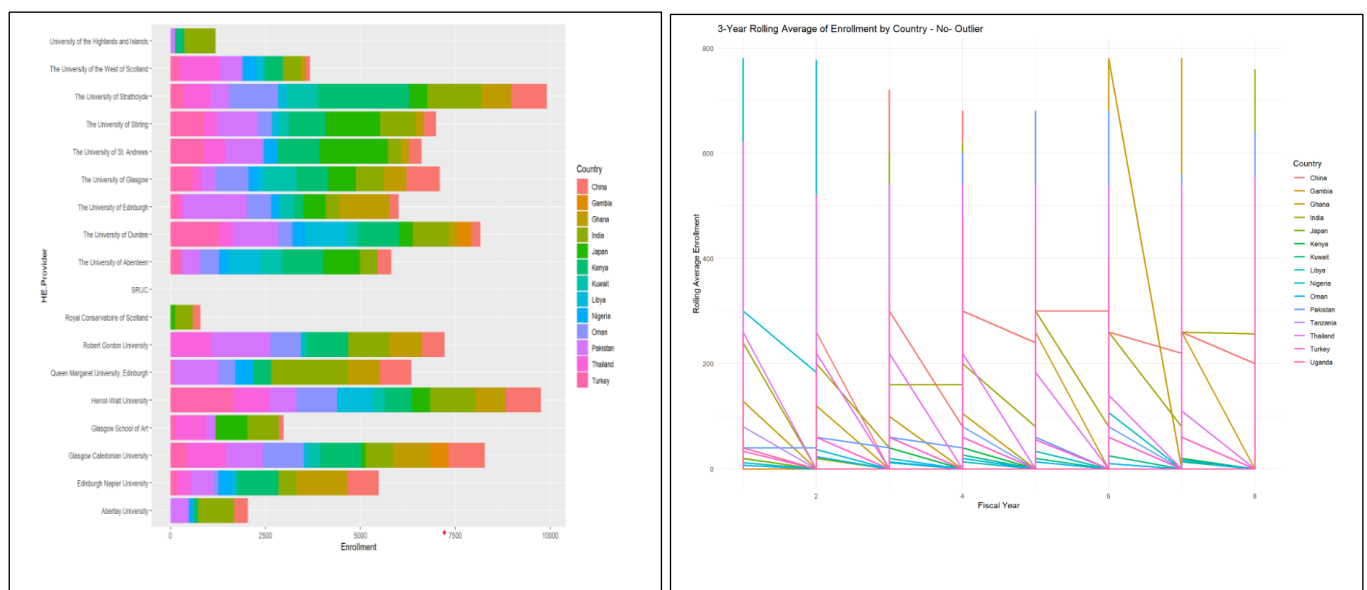


Fig 4.2: Rolling Average of Enrolment and Distribution of International Students by Country in Scottish Universities (2014-2021)

the analysis of international postgraduate student enrolment across various Scottish universities highlights distinct patterns influenced by institutional reputation and geographic location. The University of Edinburgh and the University of Glasgow emerge as the top destinations, reflecting their global standing and comprehensive educational offerings that attract students from diverse international backgrounds. The accompanying Figure (4.2), which illustrates enrolment distribution by country across different universities, reinforces this trend. Notably, China and India consistently contribute the highest number of students, aligning with broader global mobility trends where these countries are major sources of international students.

In contrast, universities like the University of the Highlands and Islands and SRUC demonstrate lower enrolment figures, suggesting either specialised program offerings or less international appeal. This discrepancy in enrolment figures indicates that while top universities benefit from established global reputations, smaller or more regionally focused institutions may need targeted recruitment strategies to increase their international student numbers. Additionally, the relatively stable enrolment figures from countries such as Japan and Oman suggest a steady demand, whereas the more fluctuating numbers from countries like Nigeria and Ghana may reflect economic or policy-related factors affecting student mobility.

This analysis aligns with global trends observed in higher education, where institutions with higher international rankings, such as the University of Edinburgh (ranked 22nd globally) and the University of Glasgow, attract more international students due to their perceived academic excellence and broader course offerings. Additionally, Scotland's appeal, supported by its high student satisfaction rates (88% of international students in Scotland are satisfied with their experience), makes it a desirable destination for international students (Jayawickrama, 2022). These trends highlight the need for Scottish universities to continuously adapt their strategies to attract and retain a diverse international student body.

4.3 Country-Specific Sensitivity to Currency Fluctuations

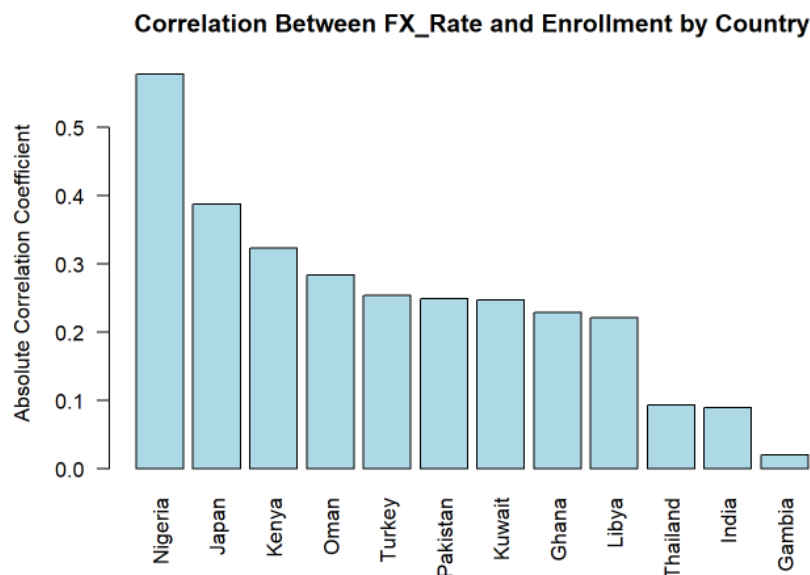


Fig 4.3: Sensitivity of International Student Enrolment to Currency Fluctuations by Country

The analysis of country-specific sensitivity to currency fluctuations, as depicted in Figure 4.3, provides critical insights into the varying impacts of exchange rate changes on international student enrolment in Scottish universities. Countries like Nigeria and Japan exhibit the highest sensitivity to fluctuations in their exchange rates with the British Pound. This is evidenced by a strong positive correlation between the depreciation of their home currencies and a corresponding decrease in student enrolment. For example, Nigeria's currency sensitivity is notably high, with a significant Pearson correlation of 0.578, indicating that as the Naira weakens, the likelihood of Nigerian students enrolling in Scottish universities decreases substantially.

This trend has broader implications for Scottish universities, as the decline in enrolment from these highly sensitive countries could push prospective students to seek education in alternative destinations such as Canada, the United States, and Germany. This shift is increasingly evident, as reported by ICEF Monitor (2024), which notes a significant decline in Nigerian student searches for study abroad opportunities in the UK, with other countries potentially benefiting from this trend.

In contrast, mid-sensitive countries such as India and China show moderate correlations with exchange rates, suggesting that while currency fluctuations do impact enrolment, other factors such as tuition fees and economic stability play a more substantial role. These countries maintain a relatively stable enrolment pattern, even amidst varying exchange rates, which could be attributed to stronger economic conditions or additional factors like scholarship opportunities or government support.

4.5 Predictive Modelling of Enrolment Based on Currency Volatility

Predictive modelling of student enrolment based on currency volatility has shown promising results, particularly when incorporating advanced machine learning techniques. The Random Forest model, which identified key predictors such as rolling averages of FX rates and economic indicators, is supported by various studies. For instance, Akinode and Bada (2021) demonstrated that decision trees and Random Forest models are highly effective in predicting student enrolment due to their ability to handle complex, non-linear relationships. Specifically, the rolling average of FX rates showed a %IncMSE of 62.58 and an IncNodePurity of 3,119,284.000, making it the most influential predictor. The FX rate itself also had a significant impact, with a %IncMSE of 36.57 and an IncNodePurity of 1,381,153.310, highlighting its role in shaping enrolment trends.

However, the current model's low R-squared value suggests that while these predictors are important, they do not fully capture the complexity of student decision-making. To address this, integrating more sophisticated techniques, such as Support Vector Machines (SVM) or Recurrent Neural Networks (RNNs), could be beneficial. SVMs have shown great potential in handling high-dimensional data, making them suitable for capturing the nuanced impacts of multiple economic indicators on enrolment (Akinode & Bada, 2021). Similarly, RNNs and Long Short-Term Memory (LSTM) networks are highly effective for time series forecasting, as they can retain information over long sequences, making them ideal for modelling the temporal dependencies inherent in enrolment data (Fong et al., 2009).

For future work, leveraging ensemble methods like Gradient Boosting or a hybrid model approach that combines ARIMA with machine learning techniques could further enhance predictive accuracy. Ensemble methods, which aggregate multiple models, have been shown to improve performance and robustness, particularly in complex scenarios such as predicting student enrolment amid fluctuating economic conditions (Shilbayeh & Abonamah, 2021). This approach, along with the inclusion of socio-economic indicators and real-time data, would likely yield more accurate and actionable insights, providing universities with better tools to anticipate and respond to changes in international student enrolment patterns.

4. 6 Visa Policy changes and trends; Impact on Enrolment

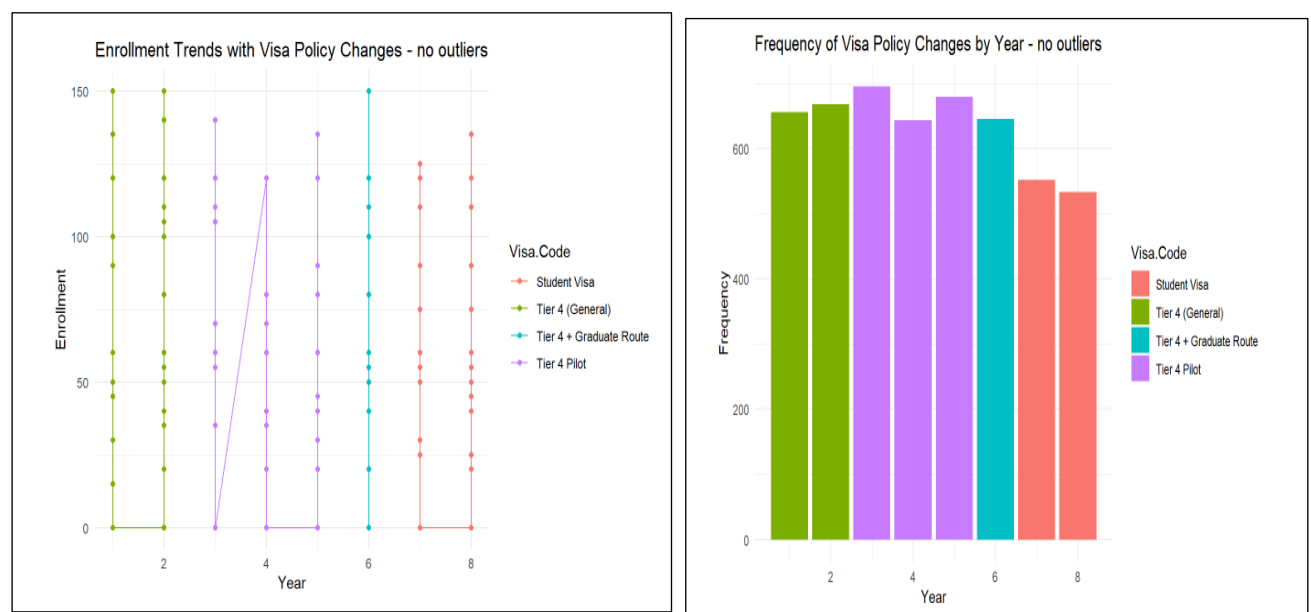


Fig 4.6: Frequency and Enrolment Trends of UK Visa Policy Changes by Year (2014-2022)

Over the years, UK student visas have undergone several name changes and policy modifications, which have directly influenced international student enrolment trends. These visas, including the Tier 4 General, Tier 4 Pilot, and the more recent Graduate Route, have all been mechanisms through which the UK has managed international student mobility. Each iteration of these visas has introduced different benefits and restrictions that have either bolstered or dampened enrolment numbers.

As depicted in Figures 4.6, the stability of certain visa categories like the Student Visa and Tier 4 General has been crucial in maintaining steady enrolment figures. The introduction of the Tier 4 Pilot in 2016 saw an initial surge in enrolments, as it provided more favourable post-study work options, which attracted a significant number of students. However, as the novelty of the Tier 4 Pilot wore off, enrolment figures stabilised, indicating that while new visa routes can create temporary spikes in interest, long-term trends depend on the sustained attractiveness of the benefits offered.

A significant spike in enrolment was observed with the introduction of the Graduate Route in 2019 under the "Student Visa" category, which allowed graduates to remain in the UK for two years post-graduation. This visa route not only increased the appeal of studying in the UK but also catered to the growing demand for post-study work opportunities, further boosting enrolment numbers.

The shift towards increased stringency in visa policies, particularly from 2023 onwards, has included higher visa fees, a rise in the NHS surcharge, and restrictions on bringing dependents. These changes, though not yet causing drastic declines in overall enrolment, have raised concerns that they may deter students from lower-income backgrounds or push potential applicants to consider other countries like Canada, the US, and Germany, which offer more favourable visa conditions (ICEF Monitor, 2024).

Despite these challenges, the overall consistency in visa policy application has meant that the UK continues to attract a steady stream of international students, albeit with fluctuations tied to specific visa categories. Moving forward, UK universities may need to advocate for more student-friendly visa policies or offer enhanced support to counteract the financial and regulatory burdens that could otherwise redirect potential students to other countries.

Chapter 5: Summary & Conclusion

5.1 Summary & Conclusion

This study has provided a comprehensive analysis of the impact of currency fluctuations on international postgraduate student enrolment in Scottish universities. The findings indicate that while economic factors such as exchange rates and GDP growth exert some influence on enrolment patterns, their effect is relatively modest. The analysis reveals that other determinants, including visa policies, tuition fees, and institutional support mechanisms, have a more pronounced impact on students' decisions. The study demonstrates that although currency volatility is a factor of consideration, it is not the predominant driver of enrolment trends. The predictive models and statistical analyses employed in this research have yielded insights that can inform strategic planning in universities, particularly in the identification of at-risk student demographics and the optimization of recruitment strategies.

5.2 Policy and Actionable Recommendations for Universities and Government

Flexible Tuition Pricing: Universities should consider adopting adaptable tuition fee structures that can accommodate fluctuations in currency exchange rates. Strategies such as offering tuition in students' home currencies or providing early payment discounts could mitigate the financial uncertainty faced by students from economically volatile regions. These measures could enhance the attractiveness of Scottish universities to international students facing financial constraints.

Enhanced Financial Aid Programs: The study underscores the necessity for universities to strengthen their financial aid offerings, particularly for students from emerging markets. Developing scholarship programs that account for currency depreciation and inflation could increase accessibility for students from these regions. Additionally, establishing emergency funds to support students experiencing unexpected financial

difficulties due to currency volatility could improve retention rates and academic success.

Targeted Recruitment: Universities should utilise the insights gained from this research to focus recruitment efforts on countries with stable economic growth and less currency volatility. However, to maintain a diverse student body, institutions should develop alternative recruitment strategies for regions experiencing economic instability, such as partnerships with local governments or educational institutions that provide financial assistance to students.

Visa and Immigration Policies: A collaborative approach between universities and government bodies is essential to ensure that visa policies remain flexible and supportive of international students, particularly about post-study work opportunities. It is recommended that the government consider revising visa policies to enhance their appeal to students from emerging markets, potentially through incentives for those pursuing studies in fields that are in high demand.

Leveraging MPower Financing and Similar Initiatives: Financial institutions such as MPower Financing, which offer loans to international students without requiring co-signers or credit history, should be promoted as viable financial solutions. These institutions play a crucial role in reducing the financial barriers for students from economically volatile regions, thereby making Scotland a more attractive destination for higher education (Mpower Financing, 2024). MPower Financing has facilitated the enrolment of students from emerging markets in the United States and Canada, contributing to the growth of these countries as preferred study destinations.

5.3 Assumptions & Limitations of the Study

This research is subject to several assumptions and limitations. It was assumed that secondary data sources, such as those from HESA and financial databases, are reliable and comprehensive. Additionally, the study presumed that currency fluctuations have a direct impact on student decision-making processes without significant delays. The

selection of variables, including GDP, inflation, and visa policies, was based on their perceived relevance to predicting enrolment trends.

The reliance on secondary data due to time constraints means that some nuanced factors influencing student decisions, such as personal or cultural reasons, may not have been captured. The exclusion of primary data collection limits the depth of the analysis. Moreover, the predictive models employed were constrained by the available data and may not fully account for future economic shifts or policy changes. The ARIMA model, though useful, showed limitations in addressing the noise and variability of the data, suggesting that future studies might benefit from more advanced or hybrid predictive techniques.

5.3 Future Research

Future research should build upon the findings of this study by integrating more advanced and diverse methodologies, particularly around model development. Firstly, the inclusion of primary data, such as surveys and interviews with international students, could provide deeper insights into the personal, cultural, and motivational factors influencing enrolment decisions, which were not fully captured by the secondary data used in this study. This approach aligns with the suggestion by Aisling (2023), who highlighted the importance of considering broader socio-cultural influences that are often overlooked in purely economic analyses.

In terms of predictive modelling, future studies should explore advanced techniques such as ensemble learning methods (e.g., Random Forest, Gradient Boosting Machines) and deep learning models like Long Short-Term Memory (LSTM) networks. These models can address the limitations of traditional linear models by capturing complex, non-linear relationships within the data. The effectiveness of such models in educational contexts has been supported by studies like those of Vanitha and Jayashree (2022), who found that ensemble methods significantly enhance predictive accuracy. Additionally, hybrid

approaches that combine ARIMA with machine learning techniques, as suggested by Fong et al. (2009), could provide more robust forecasts by leveraging the strengths of multiple methods.

Future research should also consider incorporating a broader range of predictors, including variables related to political stability, quality of education rankings, and real-time data sources like social media sentiment analysis. This approach is supported by the push-pull framework discussed by Cheng et al. (2020), which emphasises the multifaceted nature of factors influencing international student mobility. Furthermore, expanding feature engineering to include time-lagged variables and dimensionality reduction techniques could enhance the interpretability and predictive power of models, as highlighted by the need for more comprehensive models in predicting student enrolment.

A longitudinal approach using dynamic panel data models, such as the Generalised Method of Moments (GMM), could offer insights into how relationships between economic factors and enrolment evolve over time. This approach would address some of the limitations of the cross-sectional design used in this study and is consistent with the recommendations of De Haas et al. (2020), who emphasised the importance of considering temporal dynamics in migration studies. Additionally, expanding the geographical scope to include comparative studies across multiple countries would provide a more global analysis of student mobility trends, helping to understand Scotland's position within the broader international education market.

Finally, the integration of real-time data and the development of adaptive modelling techniques could enable universities to respond more effectively to economic changes, thereby enhancing their recruitment strategies in a rapidly evolving global education landscape.

By pursuing these avenues, future research can significantly advance the understanding of the complex interplay between economic factors and international postgraduate student enrolment, providing more robust and actionable insights for universities and policymakers.

Chapter 6: References

1. Abiddin, N.Z. and Ismail, A., 2011. **Attrition and completion issues in postgraduate studies for student development.** *Journal*, 1(1), pp.15–29. Available at: <http://dx.doi.org/> [Accessed 29 July 2024].
2. Akanwa, E.E., 2015. **International students in Western developed countries: History, challenges, and prospects.** *Journal of International Students*, 5(3), pp.271-284. Available at: <https://doi.org/10.32674/jis.v5i3.421>.
3. Arrowsmith, J., 2023. **UUKI responds to media speculation regarding international students.** [online] Universities UK. Available at: <https://www.universitiesuk.ac.uk/universities-uk-international/events-and-news/uuki-news/uuki-responds-media-speculation> [Accessed 6 August 2024].
4. Asogwaa, D.C., n.d. **Student attrition prediction using machine learning techniques.** *Journal*, [online] Available at: <URL>.
5. Bennett, M., 2024. **UK Masters fees for international students in 2024.** [online] FindAMasters. Available at: <https://www.findamasters.com/guides/international-postgraduate-fees> [Accessed 5 August 2024].
6. Bernard, A. and Bell, M., 2018. **Internal migration and education: A cross-national comparison.** *Cornell University*. Available at: <https://doi.org/10.48550/arxiv.1812.08913>.
7. Bradley, S. and Migali, G., 2013. **University dropouts: An evaluation of the effects of a tuition fee reform in the UK.** *Journal*, [online] Available at: <URL>.
8. Cheng, B., Lin, L. and Fan, A., 2020. **Trends, reasons, and impacts of international student mobility: A Chinese perspective.** *Springer Nature*, pp.1-19. Available at: https://doi.org/10.1007/978-981-15-5588-6_1.
9. Chevalier, A., 2022. **How to attract international students?** *IZA World of Labor*. Available at: <https://doi.org/10.15185/izawol.36.v2>.
10. Coneyworth, L., et al., 2020. **The overlooked cohort? – Improving the taught postgraduate student experience in higher education.** *Innovations in Education and Teaching International*, 57(3), pp.262-273. Available at: <https://doi.org/10.1080/14703297.2019.1617184>.
11. ICEF Monitor, 2024 Data shows a decline in Nigerian student searches for study abroad; UK may be hardest-hit, 2024. [online] - Market Intelligence for International Student Recruitment. Available at: <https://monitor.icef.com/2024/05/data-shows-a-decline-in-nigerian-student-searches-for-study-abroad-uk-may-be-hardest-hit/> [Accessed 29 July 2024].
12. Darin Wohlgemuth, D., Whalen, J., Sullivan, J., Nading Mack Shelley, C. and Wang, Y.R., 2006-2007. **Financial, academic, and environmental influences on the retention and graduation of students.** *Journal of College Student Retention: Research, Theory & Practice*, 8(4), pp.457-475.
13. Davey, K.A., 2023. **Internationalisation of the curriculum in health programs.** *BMC Medical Education*, 23(1). Available at: <https://doi.org/10.1186/s12909-023-04271-8>.

14. Delen, D., 2011. **Predicting student attrition with data mining methods.** *Journal of College Student Retention: Research, Theory & Practice*, 13(1), pp.17-35. Available at: <https://doi.org/10.2190/cs.13.1.b>.
15. Ekugo, N., 2024. **Naira devaluation: Nigerian students at Teesside University forced to discontinue studies, return home.** [online] Nairametrics. Available at: <https://nairametrics.com/2024/05/22/naira-devaluation-nigerian-students-at-teesside-university-forced-to-discontinue-studies-return-home/> [Accessed 29 July 2024].
16. Enhancing macroeconomic resilience: A comparative analysis of Nigeria and Ghana, 2024. *South African Institute of International Affairs*. Available at: <https://saiia.org.za/research/enhancing-macroeconomic-resilience-a-comparative-analysis-of-nigeria-and-ghana/>.
17. Forbes-Mewett, H. and Sawyer, A., 2016. **International students and mental health.** *Journal of International Students*, 6(3), pp.661-677. Available at: <https://doi.org/10.32674/jis.v6i3.348>.
18. Geiger, M., Trencsek, J. and Wacker, M.K., 2019. **Understanding economic growth in Ghana in comparative perspective.** *World Bank Group Policy Research Working Paper*, No. 8699. Available at: <https://doi.org/10.1596/1813-9450-8699>.
19. Ghann, P. and Abraham, Y.A., 2020. **Delivering a UK/USA program in Ghana: Opportunities, challenges and risk mitigating strategies.** *Journal*, pp.165-173. Available at: <https://doi.org/10.47987/rwzc8898>.
20. Gkontzis, A.F., et al., 2022. **A predictive analytics framework as a countermeasure for attrition of students.** *Interactive Learning Environments*, 30(6), pp.1028-1043. Available at: <https://doi.org/10.1080/10494820.2019.1709209>.
21. Glandorf, D., et al., 2024. **Temporal and between-group variability in college dropout prediction.** In: *Proceedings of the 14th Learning Analytics and Knowledge Conference*. New York, NY, USA: Association for Computing Machinery, pp.486-497. Available at: <https://doi.org/10.1145/3636555.3636906> [Accessed 29 July 2024].
22. Grebennikov, L. and Shah, M., 2012. **Investigating attrition trends in order to improve student retention.** *Quality Assurance in Education*, 20(3), pp.223-236. Available at: <https://doi.org/10.1108/09684881211240295>.
23. HESA, 2020. **Higher education student statistics: UK, 2018/19.** [online] Higher Education Statistics Agency. Available at: <https://www.hesa.ac.uk/news/16-01-2020/sb255-higher-education-student-statistics> [Accessed 8 June 2024].
24. HESA, 2024. **Table 6 - Tuition fees and education contracts analysed by HE provider, domicile, mode, level, source and academic year 2016/17 to 2022/23.** [online] Higher Education Statistics Agency (HESA). Available at: <https://www.hesa.ac.uk/data-and-analysis/finances/table-6> [Accessed 8 April 2024].
25. Herzog, S., 2005. **Measuring determinants of student return vs. dropout/stopout vs. transfer: A first-to-second year analysis of new freshmen.** *Research in Higher Education*, 46(8), pp.883-928. Available at: <https://doi.org/10.1007/s11162-005-6933-7>.
26. House, G., 2020. **Postgraduate education in the UK.** *Higher Educational Policy Institute (HEPI)*.

27. Hyams-Ssekasi, D., Mushibwe, C. and Caldwell, E., 2014. **International education in the United Kingdom**. *SAGE Open*, 4(4), pp.1-13. Available at: <https://doi.org/10.1177/2158244014562386>.
28. ICEF, 2024. **Home Office data confirms downturn in UK visa issuances through Q1 2024**. *ICEF Monitor*. Available at: <https://monitor.icef.com/2024/06/home-office-data-confirms-downturn-in-uk-visa-issuances-through-q1-2024/>.
29. Idachaba, F., 2014. **Industry-academia collaborations: The Shell sabbatical model**. *Society of Petroleum Engineers*. Available at: <https://doi.org/10.2118/172472-ms>.
30. International Student Statistics in UK 2022, 2023. *Studying-in-UK.org*. Available at: <https://www.studying-in-uk.org/international-student-statistics-in-uk/>.
31. Jayawickrama, K., 2022. **Top universities in Scotland**. *TopUniversities*. Available at: <https://www.topuniversities.com/where-to-study/europe/united-kingdom/top-universities-scotland> [Accessed 29 August 2024].
32. Kerby, M.B., 2015. **Toward a new predictive model of student retention in higher education: An application of classical sociological theory**. *Journal of College Student Retention: Research, Theory & Practice*, 17(2), pp.138-161. Available at: <https://doi.org/10.1177/1521025115578229>.
33. Kirloskar, P. and Inamdar, N., 2021. **Shifting international student mobility directions and factors influencing students' higher education destination choices**. *Journal of Higher Education Policy and Leadership Studies*, 2(3), pp.160-178. Available at: <https://doi.org/10.52547/johepal.2.3.160>.
34. Kovacic, Z., 2010. **Early prediction of student success: Mining students enrolment data**. *Journal of Information Technology Education*, 9, pp.647-665. Available at: <https://www.informingscience.org/publications/1281>.
35. Lassibille, G. and Navarro Gómez, L., 2008. **Why do higher education students drop out? Evidence from Spain**. *Education Economics*, 16(1), pp.89-105. Available at: <https://doi.org/10.1080/09645290701523267>.
36. MAC, 2018. **Impact of international students in the UK**. *Migration Advisory Committee*. Available at: <https://www.gov.uk/government/organisations/migration-advisory-committee>.
37. Maringe, F. and Carter, S., 2007. **International students' motivations for studying in UK HE: Insights into the choice and decision making of African students**. *International Journal of Educational Management*, 21(6), pp.459-475. Available at: <https://doi.org/10.1108/09513540710780000>.
38. Marshal, I. and Dumini, I.S., 2017. **Nigeria economy and the politics of recession: A critique**. *Journal of Asian Economic and Financial Review*, 2(4). Available at: <https://doi.org/10.22606/jaef.2017.24006>.
39. Massey, S.D., et al., 1993. **Theories of international migration: A review and appraisal**. *Population and Development Review*, 19(3), pp.431-466. Available at: <https://doi.org/10.2307/2938462>.
40. Meharg, D. and Craighill, S., 2014. **Student beliefs as barriers to articulation**. *Proceedings of the 44th Annual Frontiers in Education Conference (FIE)*. Available at: <https://doi.org/10.1109/fie.2014.7044416>.

41. Mok, H.K., Xiong, W. and Ke, G., 2022. **Reimagining higher education in the post-COVID-19 era: Chinese students' desires for overseas learning and implications for university governance.** *Globalisation, Societies and Education*, 20(5), pp.591-609. Available at: <https://doi.org/10.1057/s41307-022-00273-1>.
42. MPOWER Financing, 2024. **International student loans.** [online] Available at: <https://www.mpowerfinancing.com/> [Accessed 29 August 2024].
43. Murray, J., 2012. **UK tuition fees for international students.** [online] Save the Student. Available at: <https://www.savethestudent.org/international-students/international-student-fees.html> [Accessed 5 August 2024].
44. Murtaugh, P., Burns, L. and Schuster, J., 1999. **Predicting the retention of university students.** *Research in Higher Education*, 40, pp.355-371. Available at: <http://link.springer.com/10.1023/A:1018755201899>.
45. Natowitz, A., 1995. **International students in U.S. academic libraries: Recent concerns and trends.** *Research Strategies*, 13, pp.4-16.
46. Okoye, K., et al., 2024. **Machine learning model (RG-DMML) and ensemble algorithm for prediction of students' retention and graduation in education.** *Computers and Education: Artificial Intelligence*, 6, p.100205. Available at: <https://www.sciencedirect.com/science/article/pii/S2666920X24000067>.
47. Pawar, K.S., et al., 2020. **International students' motivations for studying in Indian universities: Implications for value proposition and positioning strategies.** *SAGE Open*, 15(2), pp.158-170. Available at: <https://doi.org/10.1177/1745499920910579>.
48. Pinna, M., Chiappa, D.G. and Gallarza, M.G., 2018. **Factors and information sources influencing students' consumer behaviour: A case study at an Italian public university.** *International Journal of Management in Education*, 12(4), pp.351-351. Available at: <https://doi.org/10.1504/IJMIE.2018.095160>.
49. Playdon, J., 2018. **How much does it cost to study in the UK?** [online] Top Universities. Available at: <https://www.topuniversities.com/student-info/student-finance/how-much-does-it-cost-study-uk> [Accessed 5 August 2024].
50. Praseres, L., 2013. **International and intra-national student mobility: Trends, motivations and identity.** *Geography Compass*, 7(11), pp.804-820. Available at: <https://doi.org/10.1111/gec3.12080>.
51. Qi, D., et al., 2022. **An empirical analysis of the impact of higher education on economic growth: The case of China.** *Frontiers in Psychology*, 13. Available at: <https://doi.org/10.3389/fpsyg.2022.959026>.
52. Schmitt, N., et al., 2009. **Prediction of 4-year college student performance using cognitive and noncognitive predictors and the impact on demographic status of admitted students.** *Journal of Applied Psychology*, 94(6), pp.1479-1497. Available at: <http://dx.doi.org/10.1037/a0016810>.
53. Scotland, Q.A.A., 2017. **Review (ELIR) reports 2013-16: Transitions into university.** *The Quality Assurance Agency for Higher Education*.
54. Scottish Government, 2024. **Scotland's international education strategy.** *Scottish Government*.

55. Scottish Government, 2023. **More Scottish students than ever in Scotland's universities.** [online] Available at: <https://www.gov.scot/news/more-scottish-students-than-ever-in-scotlands-universities/> [Accessed 6 August 2024].
56. Shayah, H.M. and Sun, Z., 2019. **Employment in the Gulf Cooperation Council (GCC) countries – Current issues and future trends.** *Proceedings of the 7th International Conference on Social Science and Higher Education (ICSSHE 2018)*. Available at: <https://doi.org/10.2991/ssphe-18.2019.94>.
57. Shkoler, O. and Rabenu, E., 2022. **The motivations and their conditions which drive students to seek higher education in a foreign country.** *Current Psychology*, 42(29), pp.25403-25416. Available at: <https://doi.org/10.1007/s12144-022-03619-5>.
58. Soo, K.T. and Elliott, C., 2010. **Does price matter? Overseas students in UK higher education.** *Economics of Education Review*, 29(4), pp.553-565. Available at: <https://www.sciencedirect.com/science/article/pii/S027277570900140X>.
59. Sturrock, D. and O'dea, C., 2019. **Survival pessimism and the demand for annuities.** *The Review of Economics and Statistics*, 105, pp.442-457. Available at: <https://direct.mit.edu/rest/article/105/2/442/100992/survival-pessimism-and-the-demand-for-annuities>.
60. Student Loans Company, 2020. **Student support for higher education in England 2020.** [online] Available at: <https://www.gov.uk/government/statistics/student-support-for-higher-education-in-england-2020> [Accessed 6 August 2024].
61. Tarmizi, S.S.A., et al., 2019. **A review on student attrition in higher education using big data analytics and data mining techniques.** *International Journal of Modern Education and Computer Science*, 11(8), pp.1-14. Available at: https://www.researchgate.net/publication/337896368_a_review_on_student_attrition_in_higher_education_using_big_data_analytics_and_data_mining_techniques [Accessed 29 July 2024].
62. Taylor, G. and Ali, N., 2017. **Learning and living overseas: Exploring factors that influence meaningful learning and assimilation: How international students adjust to studying in the UK from a socio-cultural perspective.** *Education Sciences*, 7(1), p.35. Available at: <https://doi.org/10.3390/educsci7010035>.
63. The Impact of Financial Support on Student Experience, 2019. [online] National Union of Students. Available at: <https://www.nus.org.uk/en/news/the-impact-of-financial-support-on-student-experience/> [Accessed 2024].
64. The Scottish Government, 2021. **The impact of international students in Scotland: Scottish Government response to the Migration Advisory Committee's consultation on the impact of international students in the UK.** *The Scottish Government*.
65. Vaarma, M. and Li, H., 2024. **Predicting student dropouts with machine learning: An empirical study in Finnish higher education.** *Technology in Society*, 76, p.102474. Available at: <https://www.sciencedirect.com/science/article/pii/S0160791X24000228>.
66. Weber, T. and Mol, V.C., 2023. **The student migration transition: An empirical investigation into the nexus between development and international student migration.** *Comparative Migration Studies*, 11(1). Available at: <https://doi.org/10.1186/s40878-023-00329-0>.

67. Xiao, W., Ji, P. and Hu, J., 2022. **A survey on educational data mining methods used for predicting students' performance.** *Engineering Reports*, 4(5). Available at: <https://onlinelibrary.wiley.com/doi/10.1002/eng2.12482>.
68. Yağci, M., 2022. **Educational data mining: Prediction of students' academic performance using machine learning algorithms.** *Smart Learning Environments*, 9(1), pp.1-19. Available at: <https://slejournal.springeropen.com/articles/10.1186/s40561-022-00192-z> [Accessed 29 July 2024].
69. Yang, P., 2017. **Understanding 'integration': Chinese 'foreign talent' students in Singapore talking about rongru.** *The Journal of Transnational Education and Mobility*, 1(1), pp.29-45. Available at: https://doi.org/10.1386/tjtm.1.1.29_1.
70. Yukselturk, E., Osekes, S. and Türel, Y.K., 2014. **Predicting dropout student: An application of data mining methods in an online education program.** *European Journal of Open, Distance and E-Learning*, 17(1), pp.118-133. Available at: <https://sciendo.com/article/10.2478/eurodl-2014-0008> [Accessed 29 July 2024].

Chapter 7: Appendix

Appendix 1 – Project Plan Summary

Week Number/ Dates	Title	Tasks	Tools
WK1 24/6/24 30/6/24	Literature Review & Research Preparation	- Browsed for articles in related areas. - Conducted a literature review on international currency fluctuations. - Reviewed related economic theories. - Searched for valid sites with historical currency data.	Internet, Google-Scholar, RGU library, Libkey.io
WK2 /29/6/24 30/6/24	Data Collection & Initial Analysis	- Gathered historical data on currency fluctuations. - Identified key economic indicators relevant to the study. - Prepared datasets for analysis. - Began preliminary data analysis on the collected data.	Ms-Excel, R
WK3/ 08/7/24 14/7/24	Data Cleaning & Validation	- Cleaned and organised the collected data. - Validated data accuracy and consistency. - Performed additional data collection. - Documented the data cleaning process.	Ms-Excel, R
WK4/ 15/7/24 21/7/24	Exploratory Data Analysis	- Conducted in-depth analysis of currency fluctuations and their impact. - Explored correlations between currency data, visa changes, economic indicator data, tuition fee data and student enrolment figures. - Identified patterns and trends. - Prepared initial visualisations and summaries.	R
WK5/ 22/7/24 28/7/24	Model Development	- Developed forecasting models based on the analysed data. - Tested the models for accuracy and reliability. - Refined models based on test results. - Documented the model development process.	R
WK 29/7/24 4/8/24	Model Application & Analysis	- Applied the forecast models to different scenarios. - Analysed potential future trends in student enrolment based on currency fluctuations. - Compared model outputs with actual historical data. - Prepared evaluation and analysis reports.	R

WK 05/8/24 11/8/24	7/ -	Report Writing & Visualisations	• Compiled findings into a comprehensive report. • Created detailed visualisations to support the findings. • Drafted the initial sections of the report. • Reviewed and refined content for clarity and accuracy.	R, Ms-Word
WK 12/8/24 18/8/24	8/ -	Final Preparation & Submission	• Finalising the report content and visualisations. • Review and proofreading. • Prepared the report for submission. • Submitted the final report	Ms-Word

Table: 7.1 shows a summarised project roadmap

Appendix 2 – Code snippets of the Pearson Correlation on the no outlier dataset

```
## [1] "Pearson Correlation: GDP Value on Enrollment Numbers"
```

```
print(pearson_corr2)
```

```
##
## Pearson's product-moment correlation
##
## data: final_combined_1422_no_outlier$Value and final_combined_1422_no_outlier$Enrollment
## t = 2.5834, df = 5073, p-value = 0.009811
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.008741964 0.063697102
## sample estimates:
## cor
## 0.03624694
```

```
## [1] "Pearson Correlation: tuition fees on Enrollment Numbers"
```

```
print(pearson_corr3)
```

```
##
## Pearson's product-moment correlation
##
## data: final_combined_1422_no_outlier$Tuition_Av and final_combined_1422_no_outlier$Enrollment
## t = 1.6286, df = 5073, p-value = 0.1035
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.004657182 0.050341465
## sample estimates:
## cor
## 0.02285944
```

```
## [1] "Pearson Correlation: FX on Enrollment Numbers"
```

```
print(pearson_corr)
```

```
##
## Pearson's product-moment correlation
##
## data: final_combined_1422_no_outlier$FX_Rate and final_combined_1422_no_outlier$Enrollment
## t = 3.5719, df = 5073, p-value = 0.0003577
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.02260412 0.07749356
## sample estimates:
## cor
## 0.05008666
```

```
## [1] "Pearson Correlation: Visa fees on Enrollment Numbers"
```

```
print(pearson_corr4)
```

```
##
## Pearson's product-moment correlation
##
## data: final_combined_1422_no_outlier$Visa.Fee.... and final_combined_1422_no_outlier$Enrollment
## t = -2.5174, df = 5073, p-value = 0.01185
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.062775275 -0.007816499
## sample estimates:
## cor
## -0.03433750
```

Fig: 7.1 (Appendix 2)

Appendix 3- Folder content

Folder Name – Appendix